
Ricardo Leite Martins Bazarin - Academic Documentation

Ricardo Leite Martins Bazarin

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- [Curriculum Lattes \(Brazilian Academic Curriculum\)](#)
- [GitHub Page](#)
- [Google Scholar](#)

Articles Published in Periodics

- [2021-Boundary Effects on the Tortuosity and Permeability of Idealized Porous Media](#)
 - Authors: Bazarin, De Lai, Naaktgeboren, Junqueira.
 - Area: (Porous media fluid-flow) Single-phase porous properties characterization in fractal porous media
- [2021-Moments-based method for boundary conditions in the lattice Boltzmann framework: A comparative analysis for the lid driven cavity flow](#)
 - Authors: Bazarin, Philippi, Randles, Hegele Jr.
 - Area: (Boundary conditions modeling)
- [2024-Improved lattice Boltzmann model for immiscible multicomponent systems with high viscosity gradients at the interface](#)
 - Authors: Bazarin, Naaktgeboren, Junqueira, Philippi, Hegele Jr.
 - Area: (Multicomponent/multiphase modeling) Numerical modeling for multicomponent fluid-flow
- [2025-Lattice Boltzmann method assessment for the Buckley–Leverett equation: Extension to dynamic capillary effect](#)
 - Authors: Bazarin, dos Santos, Siebert.
 - Area: (Porous media fluid-flow - Multicomponent/multiphase modeling) Numerical modeling of multicomponent/multiphase flow in porous-continuous porous media.

- Supplemental Materials - [C++ code of lattice Boltzmann simulation](#) presented.

Academic Material

The full table of contents is generated automatically from the `_toc.yml` file and appears in the sidebar (HTML) or in the table of contents (PDF).

DIFFUSIVE EQUATION

The diffusive equation is described by

$$\partial_t \phi - \partial_\alpha (\nu \partial_\alpha \phi) = 0 \quad (1.1)$$

where $\phi(\alpha, t)$ is a time and space dependent parameter and ν is diffusive coefficient.

1.1 Numerical Discretization

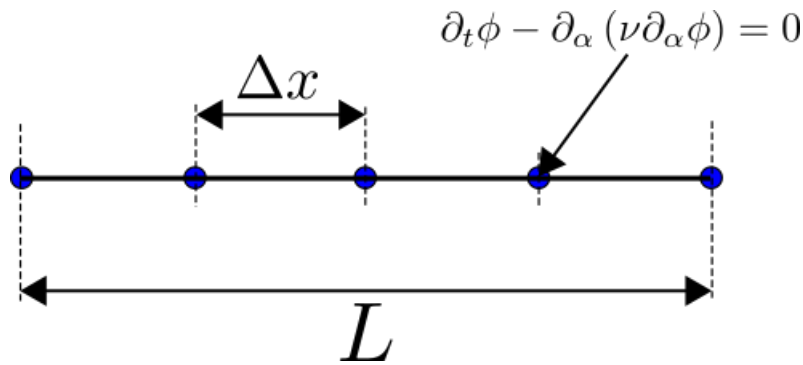


Fig. 1.1: 1D regular grid example for diffusive equation.

1.1.1 FDM Discretization

Discretizing the Eq. (1.1) through finite difference method, we have for forward time and central space (FTCS) discretizations:

$$\partial_t \phi - \partial_\alpha (\nu \partial_\alpha \phi) = \frac{\phi(x, t + \Delta_t) - \phi(x, t)}{\Delta_t} - \nu \frac{\phi(x - \Delta_x, t) - 2\phi(x, t) + \phi(x + \Delta_x, t)}{\Delta_x^2} = 0,$$

for ν constant and isolating the term $\phi(x, t + \Delta_t)$, we can describe the ϕ time evolution:

$$\phi(x, t + \Delta_t) = \phi(x, t) + \frac{\nu \Delta_t}{\Delta_x^2} (\phi(x - \Delta_x, t) - 2\phi(x, t) + \phi(x + \Delta_x, t)).$$

1.1.2 Lattice Boltzmann Equation

Describing the problem through the BGK lattice Boltzmann equation (BGK-LB):

$$f_i(x_\alpha + c_{i,\alpha}\delta t, t + \delta t) = f_i(x_\alpha, t) - \left(\frac{f_i - f_i^{eq}}{\tau} \right), \quad (1.2)$$

where the equilibrium distribution function is defined by

$$f_i^{eq} = w_i \phi(x_\alpha, t),$$

and the equilibrium moments are given by

$$\sum_{i=0} f_i^{eq} = \phi, \quad \sum_i f_i^{eq} c_{i,\alpha} = 0 \quad \text{and} \quad \sum_i f_i^{eq} c_{i,\alpha} c_{i,\beta} = c_s^2 \phi \delta_{\alpha\beta}. \quad (1.3)$$

Through Chapman–Enskog analysis, it is demonstrated that the first-order non-equilibrium moment describes the pressure gradient and, consequently, the average velocity:

$$m_\alpha^{neq} = \sum_i c_{i,\alpha} (f_i - f_i^{eq}) = -(c_s^2 \tau \delta_t) \partial_\alpha \phi \quad \text{and} \quad \tau = \frac{\nu}{c_s^2 \delta_t} + \frac{1}{2}.$$

Lattice Direction Moments

$$\begin{aligned} \underbrace{\sum_{i=0} f_i^{eq}}_{\text{Zero-Order Moment}} &= \sum_{i=0} f_i = \phi, & \underbrace{\sum_{i=0} f_i^{eq} c_{i,x}}_{\text{x-First-Order Moment}} &= 0, & \underbrace{\sum_{i=0} f_i^{eq} c_{i,y}}_{\text{y-First-Order Moment}} &= 0, \\ \underbrace{\sum_{i=0} f_i^{eq} c_{i,x} c_{i,x}}_{\text{xx-Second-Order Moment}} &= \frac{\phi}{3}, & \underbrace{\sum_{i=0} f_i^{eq} c_{i,x} c_{i,y}}_{\text{xy-Second-Order Moment}} &= 0 & \text{and} & \underbrace{\sum_{i=0} f_i^{eq} c_{i,y} c_{i,y}}_{\text{yy-Second-Order Moment}} = \frac{\phi}{3} \end{aligned}$$

Chapmann-Enskog Analysis

Applying the Chapmann-Enskog procedure to LB equation, we expand the term $f_i(x + e_i \Delta t, t + \Delta t)$ in a Taylor series to recover the derivative form of the equation, i.e.,

$$f_i(x_\alpha + c_{i,\alpha} \Delta t, t + \Delta t) - f_i(x_\alpha, t) = \sum_{j=1}^{\infty} \frac{\Delta t^j}{j!} D_t^j f_i = - \left(\frac{f_i - f_i^{eq}}{\tau} \right). \quad (1.4)$$

Rescaling the dimensionless form of the Eq. (1.4) in terms of the Knudsen number (Kn), we have

$$\sum_{j=1}^{\infty} \frac{Kn^{(j-1)}}{j!} D_t^j f_i = - \frac{1}{Kn} \left(\frac{f_i - f_{eq,i}}{\tau} \right) \quad \rightarrow \quad \sum_{j=1}^{\infty} \frac{Kn^{(j)}}{j!} D_t^j f_i = - \frac{f_i - f_i^{eq}}{\tau},$$

applying the asymptotic expansion in both the distribution function ($f_i = f_i^{(0)} + Kn f_i^{(1)} + Kn^2 f_i^{(2)} + \dots$) and time partial derivative ($\partial_t = \partial_t^{(0)} + Kn \partial_t^{(1)} + Kn^2 \partial_t^{(2)} + \dots$), and separating the equation in orders up to the order Kn^2 :

$$\begin{aligned}
 (Kn^{(0)}) : f_i^{(0)} &= f_i^{eq}, \\
 (Kn^{(1)}) : (\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha) f_i^{(0)} &= -\frac{f_i^{(1)}}{\tau}, \\
 (Kn^{(2)}) : \partial_t^{(1)} f_i^{(0)} + (\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha) f_i^{(1)} + \frac{(\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha)^2 f_i^{(0)}}{2} &= -\frac{f_i^{(2)}}{\tau}, \\
 \text{ou} \\
 (Kn^{(2)}) : \partial_t^{(1)} f_i^{(0)} + (\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha) \left(1 - \frac{1}{2\tau}\right) f_i^{(1)} &= -\frac{f_i^{(2)}}{\tau} \quad \rightarrow \quad f_i^{(1)} \text{ Formulation} \\
 \text{ou} \\
 (Kn^{(2)}) : \partial_t^{(1)} f_i^{(0)} + (\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha)^2 \left(\frac{1}{2} - \tau\right) f_i^{(0)} &= -\frac{f_i^{(2)}}{\tau} \quad \rightarrow \quad f_i^{(0)} \text{ Formulation.}
 \end{aligned} \tag{1.5}$$

Zero-Order Moment Balance

To retrieve the balance equation, we sum the Eq. (1.5) for $Kn^{(1)}$ and $Kn^{(2)}$ over $\sum_{i=0}$:

$$\begin{aligned}
 (Kn^{(1)}) : \sum_i (\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha) f_i^{(0)} &= \sum_i \left(-\frac{f_i^{(1)}}{\tau}\right), \\
 (Kn^{(1)}) : \partial_t^{(0)} \phi &= 0,
 \end{aligned}$$

and

$$\begin{aligned}
 (Kn^{(2)}) : \sum_i \left(\partial_t^{(1)} f_i^{(0)} + (\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha)^2 \left(\frac{1}{2} - \tau\right) f_i^{(0)}\right) &= \sum_i \left(-\frac{f_i^{(2)}}{\tau}\right), \\
 (Kn^{(2)}) : \partial_t^{(1)} \phi + \left(\frac{1}{2} - \tau\right) \partial_\alpha \partial_\beta (c_s^2 \phi \delta_{\alpha\beta}) &= 0.
 \end{aligned}$$

or

$$\begin{aligned}
 (Kn^{(2)}) : \sum_i \left[\partial_t^{(1)} f_i^{(0)} + \left(1 - \frac{1}{2\tau}\right) (\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha) f_i^{(1)}\right] &= \sum_i \left(-\frac{f_i^{(2)}}{\tau}\right), \\
 (Kn^{(2)}) : \partial_t^{(1)} \phi + \left(1 - \frac{1}{2\tau}\right) m_\alpha^{(1)} &= 0.
 \end{aligned}$$

where $m_\alpha^{(1)}$ is the first-order moment of $f_i^{(1)}$ and $\sum_{i=1} f_i^{(j)} = 0$ for $j \geq 1$ due to imposition of ϕ conservation. To compute the moment $m_\alpha^{(1)}$, we multiply Eq. (1.5) by $c_{i,\alpha}$ and sum over all lattice directions:

$$\begin{aligned}
 (Kn^{(1)}) : \sum_i c_{i,\alpha} (\partial_t^{(0)} + e_{\beta,i} \partial_\beta) f_i^{(0)} &= \sum_i c_{i,\alpha} \left(-\frac{f_i^{(1)}}{\tau}\right), \\
 (Kn^{(1)}) : m_\alpha^{(1)} &= -\tau \partial_\beta (c_s^2 \phi \delta_{\alpha\beta}).
 \end{aligned} \tag{1.6}$$

By substituting Eq. (1.6) into Eq. (1.6), we recover the Eq. (1.1), where $\nu = c_s^2(\tau - 1/2)$. In the regularized formulation of the lattice Boltzmann equation, the first-order correction is approximated as $f_i^{(1)} \approx (f_i - f_i^{(0)}) = f_i^{neq}$, where higher-order Knudsen moments are filtered out and the collision term is reformulated in terms of f_i^{neq} .

Boundary Conditions

The boundary conditions for the lattices can be derived by solving a linear system of known moments.

D1Q2:	Boundaries	Layers	
		West	East
	Unknown f_i	$f_2 = \phi_e - f_1$	$f_1 = \phi_w - f_2$
D1Q3:	Boundaries	Layers	
		West	East
	Unknown f_i	$f_2 = \phi_e - f_1 - f_0$	$f_1 = \phi_w - f_2 - f_0$

1.2 1st Benchmark: Single sine mode on a finite rod (Dirichlet)

PDE: $(\phi_t = \partial_x(\nu \partial_x \phi), (0 < x < L, t > 0))$

BC: $(\phi(0, t) = \phi(L, t) = 0)$

IC: $\phi(x, 0) = \sin\left(\frac{\pi x}{L}\right)$

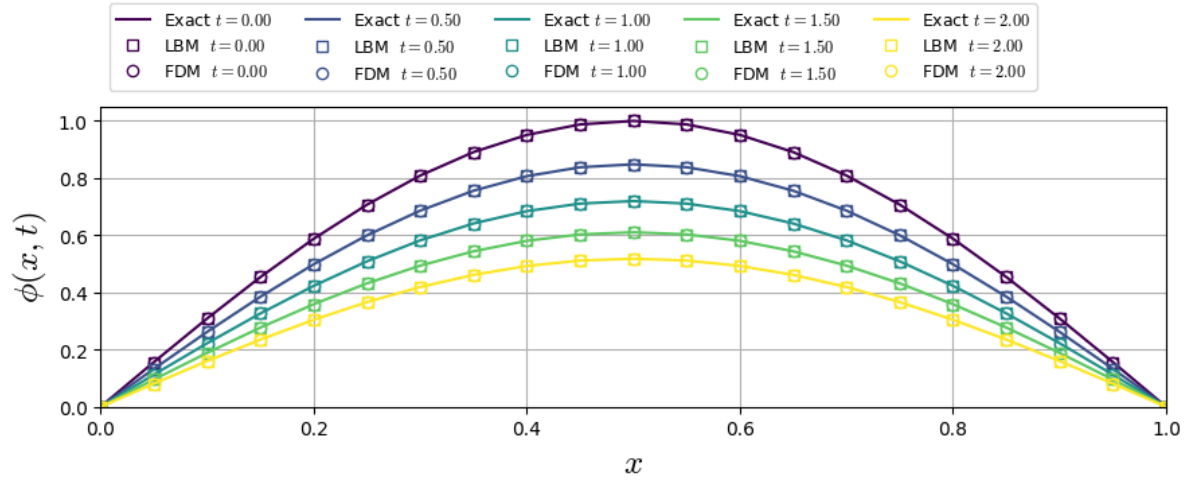
Solution

$$\phi(x, t) = \sin\left(\frac{\pi x}{L}\right) \exp\left(-\nu \frac{\pi^2}{L^2} t\right).$$

1.2.1 LBM D1Q3 Solution ($\tau = 1$)

1.2.2 FDM Solution

1.2.3 Numerical and Analytical Comparison



POISSON EQUATION (STEADY-STATE DIFFUSIVE EQUATION)

The Poisson is partial differential equation employed to represent problems as the pressure distribution over steady-state in space. It is characterized by a steady-state version of the diffusive equation, i.e, the parameter is independent of time. The partial differential equation is given by

$$\partial_{\alpha} (\nu \partial_{\alpha} p) = 0 \quad (2.1)$$

where p is the pressure and ν diffusive control parameter.

2.1 Numerical Discretization

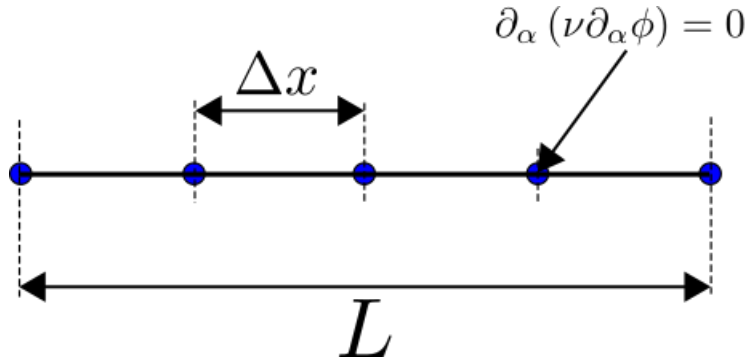


Fig. 2.1: 1D regular grid example for diffusive equation.

2.1.1 FDM Discretization

Discretizing the Eq. (1.1) through finite difference method, we have for forward time and central space (FTCS) discretizations:

$$\partial_{\alpha} (\nu \partial_{\alpha} p) = \nu \frac{p(x - \Delta_x, t) - 2p(x, t) + p(x + \Delta_x, t)}{\Delta_x^2} = 0,$$

for ν constant and isolating the term $u(x, t + \Delta_t)$, we can describe the p time evolution:

$$p(x, t) = \frac{p(x - \Delta_x, t) + p(x + \Delta_x, t)}{2}.$$

2.1.2 Lattice Boltzmann Equation

Describing the problem through the BGK lattice Boltzmann equation (BGK-LB) for the lattice D1Q3:

$$f_i(x_\alpha + c_{i,\alpha}\delta t, t' + \delta t) = f_i(x_\alpha, t') - \left(\frac{f_i - f_i^{eq}}{\tau} \right), \quad (2.2)$$

the equilibrium distribution function is defined by

$$f_i^{eq} = \begin{cases} (w_0 - 1)\hat{p}(x_\alpha, t') & i = 0 \\ w_i\hat{p}(x_\alpha, t') & i \neq 0 \end{cases},$$

where $\hat{p}$ is a shifted pressure, shifted by a reference constant pressure p_0 , i.e., $p = p_0 + \hat{p}$ (This assumption is necessary due to pressure by definition can not be negative, but negative pressure values can be found depending of initial simulation set). The equilibrium moments are given by

$$\sum_{i=0} f_i^{eq} = 0, \quad \frac{1}{1-w_0} \sum_{i=1} f_i^{eq} = \frac{1}{1-w_0} \sum_{i=1} f_i = \hat{p}, \quad \sum_i f_i^{eq} c_{i,\alpha} = 0 \quad \text{and} \quad \sum_i f_i^{eq} c_{i,\alpha} c_{i,\beta} = c_s^2 \hat{p} \delta_{\alpha\beta}. \quad (2.3)$$

Through Chapman–Enskog analysis, it is demonstrated that the first-order non-equilibrium moment describes the pressure gradient and, consequently, the average velocity:

$$m_\alpha^{neq} = \sum_i c_{i,\alpha} (f_i - f_i^{eq}) = -(c_s^2 \tau \delta_t) \partial_\alpha \hat{p} \quad \text{and} \quad \tau = \frac{\nu}{c_s^2 \delta_t} + \frac{1}{2}.$$

Lattice Direction Moments

$$\begin{aligned} \underbrace{\sum_{i=0} f_i^{eq} = \sum_{i=0} f_i = 0}_{\text{Zero-Order Moment}}, \quad & \underbrace{\frac{1}{1-w_0} \sum_{i=1} f_i^{eq} = \frac{1}{1-w_0} \sum_{i=1} f_i = \hat{p}}_{\text{Zero-Order Moment}}, \quad \underbrace{\sum_{i=0} f_i^{eq} c_{i,x} = 0}_{\text{x-First-Order Moment}}, \quad \underbrace{\sum_{i=0} f_i^{eq} c_{i,y} = 0}_{\text{y-First-Order Moment}} \\ \underbrace{\sum_{i=0} f_i^{eq} c_{i,x} c_{i,x} = \frac{\hat{p}}{3}}_{\text{xx-Second-Order Moment}}, \quad & \underbrace{\sum_{i=0} f_i^{eq} c_{i,x} c_{i,y} = 0}_{\text{xy-Second-Order Moment}} \quad \text{and} \quad \underbrace{\sum_{i=0} f_i^{eq} c_{i,y} c_{i,y} = \frac{\hat{p}}{3}}_{\text{yy-Second-Order Moment}} \end{aligned}$$

Chapmann-Enskog Analysis

Applying the Chapmann-Enskog procedure to LB equation, we expand the term $f_i(x + e_i \Delta t, t + \Delta t)$ in a Taylor series to recover the derivative form of the equation, i.e.,

$$f_i(x_\alpha + c_{i,\alpha} \Delta t, t + \Delta t) - f_i(x, t) = \sum_{j=1}^{\infty} \frac{\Delta t^j}{j!} D_t^j f_i = - \left(\frac{f_i - f_i^{eq}}{\tau} \right). \quad (2.4)$$

Rescaling the dimensionless form of the Eq. (2.4) in terms of the Knudsen number (Kn), we have

$$\sum_{j=1}^{\infty} \frac{Kn^{(j-1)}}{j!} D_t^j f_i = - \frac{1}{Kn} \left(\frac{f_i - f_{eq,i}}{\tau} \right) \quad \rightarrow \quad \sum_{j=1}^{\infty} \frac{Kn^{(j)}}{j!} D_t^j f_i = - \frac{f_i - f_i^{eq}}{\tau},$$

applying the asymptotic expansion in both the distribution function ($f_i = f_i^{(0)} + Kn f_i^{(1)} + Kn^2 f_i^{(2)} + \dots$) and time partial derivative ($\partial_t = \partial_t^{(0)} + Kn \partial_t^{(1)} + Kn^2 \partial_t^{(2)} + \dots$), and separating the equation in orders up to the order Kn^2 :

$$\begin{aligned}
 (Kn^{(0)}) : \quad & f_i^{(0)} = f_i^{eq}, \\
 (Kn^{(1)}) : \quad & (\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha) f_i^{(0)} = -\frac{f_i^{(1)}}{\tau}, \\
 (Kn^{(2)}) : \quad & \partial_t^{(1)} f_i^{(0)} + (\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha) f_i^{(0)} + \frac{(\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha)^2 f_i^{(0)}}{2} = -\frac{f_i^{(2)}}{\tau}, \\
 \text{ou} \\
 (Kn^{(2)}) : \quad & \partial_t^{(1)} f_i^{(0)} + (\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha) \left(1 - \frac{1}{2\tau}\right) f_i^{(1)} = -\frac{f_i^{(2)}}{\tau} \quad \rightarrow \quad f_i^{(1)} \text{ Formulation} \\
 \text{ou} \\
 (Kn^{(2)}) : \quad & \partial_t^{(1)} f_i^{(0)} + (\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha)^2 \left(\frac{1}{2} - \tau\right) f_i^{(0)} = -\frac{f_i^{(2)}}{\tau} \quad \rightarrow \quad f_i^{(0)} \text{ Formulation.}
 \end{aligned} \tag{2.5}$$

Zero-Order Moment Balance

To retrieve the balance equation, we sum the Eq. (2.5) for $Kn^{(1)}$ and $Kn^{(2)}$ over $\sum_{i=0}$:

$$\begin{aligned}
 (Kn^{(1)}) : \quad & \sum_i (\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha) f_i^{(0)} = \sum_i \left(-\frac{f_i^{(1)}}{\tau}\right), \\
 (Kn^{(1)}) : \quad & 0 = 0,
 \end{aligned}$$

and

$$\begin{aligned}
 (Kn^{(2)}) : \quad & \sum_i \left(\partial_t^{(1)} f_i^{(0)} + (\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha)^2 \left(\frac{1}{2} - \tau\right) f_i^{(0)} \right) = \sum_i \left(-\frac{f_i^{(2)}}{\tau}\right), \\
 (Kn^{(2)}) : \quad & \left(\frac{1}{2} - \tau\right) \partial_\alpha \partial_\beta (c_s^2 \hat{p} \delta_{\alpha\beta}) = 0.
 \end{aligned}$$

or

$$\begin{aligned}
 (Kn^{(2)}) : \quad & \sum_i \left[\partial_t^{(1)} f_i^{(0)} - \left(1 - \frac{1}{2\tau}\right) (\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha) f_i^{(1)} \right] = \sum_i \left(-\frac{f_i^{(2)}}{\tau}\right), \\
 (Kn^{(2)}) : \quad & -\left(1 - \frac{1}{2\tau}\right) m_\alpha^{(1)} = 0.
 \end{aligned}$$

where $m_\alpha^{(1)}$ is the first-order moment of $f_i^{(1)}$ and $1/(1-w_0) \sum_{i=1} f_i^{(j)} = 0$ for $j \geq 1$ due to imposition of $\hat{p}$ conservation.

To compute the moment $m_\alpha^{(1)}$, we multiply Eq. (2.5) by $c_{i,\alpha}$ and sum over all lattice directions:

$$\begin{aligned}
 (Kn^{(1)}) : \quad & \sum_i c_{i,\alpha} (\partial_t^{(0)} + e_{\beta,i} \partial_\beta) f_i^{(0)} = \sum_i c_{i,\alpha} \left(-\frac{f_i^{(1)}}{\tau}\right), \\
 (Kn^{(1)}) : \quad & m_\alpha^{(1)} = -\tau \partial_\beta (c_s^2 \hat{p} \delta_{\alpha\beta}).
 \end{aligned} \tag{2.6}$$

By substituting Eq. (2.6) into Eq. (2.6), we recover the Eq. (2.1), where $\nu = c_s^2(\tau - 1/2)$. In the regularized formulation of the lattice Boltzmann equation, the first-order correction is approximated as $f_i^{(1)} \approx (f_i - f_i^{(0)}) = f_i^{neq}$, where higher-order Knudsen moments are filtered out and the collision term is reformulated in terms of f_i^{neq} .

Boundary Conditions

The boundary conditions for the lattice D2Q5 can be derived by solving a linear system of known moments, as illustrated in appendix/bcs-D2Q5-poisson-equation. For the D2Q5 lattice, the unknown distribution functions at each boundary are determined as summarized in Table below.

Boundaries	Layers	
	North	South
Unknown f_i	$f_2 = \hat{p}_n(1 - w_0) - (f_1 + f_3 + f_4)$	$f_4 = \hat{p}_s(1 - w_0) - (f_1 + f_2 + f_3)$
	East	West
	$f_3 = \hat{p}_e(1 - w_0) - (f_1 + f_2 + f_4)$	$f_1 = \hat{p}_w(1 - w_0) - (f_2 + f_3 + f_4)$
Boundaries	Concave Corners	
	Northwest	Northeast
Unknown f_i	$f_1 = -(\partial_x \hat{p} + \partial_y \hat{p})\tau c_s^2/2 - f_2 + \hat{p}_{nw}/3$	$f_3 = (\partial_x \hat{p} - \partial_y \hat{p})\tau c_s^2/2 - f_2 + \hat{p}_{ne}/3$
	$f_4 = (\partial_x \hat{p} + \partial_y \hat{p})\tau c_s^2/2 - f_3 + \hat{p}_{nw}/3$	$f_4 = -(\partial_x \hat{p} - \partial_y \hat{p})\tau c_s^2/2 - f_1 + \hat{p}_{ne}/3$
	Southwest	Southeast
	$f_1 = -(\partial_x \hat{p} - \partial_y \hat{p})\tau c_s^2/2 - f_4 + \hat{p}_{sw}/3$	$f_2 = -(\partial_x \hat{p} + \partial_y \hat{p})\tau c_s^2/2 - f_1 + \hat{p}_{se}/3$
	$f_2 = (\partial_x \hat{p} - \partial_y \hat{p})\tau c_s^2/2 - f_3 + \hat{p}_{sw}/3$	$f_3 = (\partial_x \hat{p} + \partial_y \hat{p})\tau c_s^2/2 - f_4 + \hat{p}_{se}/3$

2.2 1st Benchmark: Single sine mode on a finite rod (Dirichlet)

PDE: $(\phi_t = \partial_x(\nu \partial_x \phi), (0 < x < L, t > 0))$

BC: $(\phi(0, t) = \phi(L, t) = 0)$

IC: $\phi(x, 0) = \sin\left(\frac{\pi x}{L}\right)$

Solution

$$\phi(x, t) = \sin\left(\frac{\pi x}{L}\right) \exp\left(-\nu \frac{\pi^2}{L^2} t\right).$$

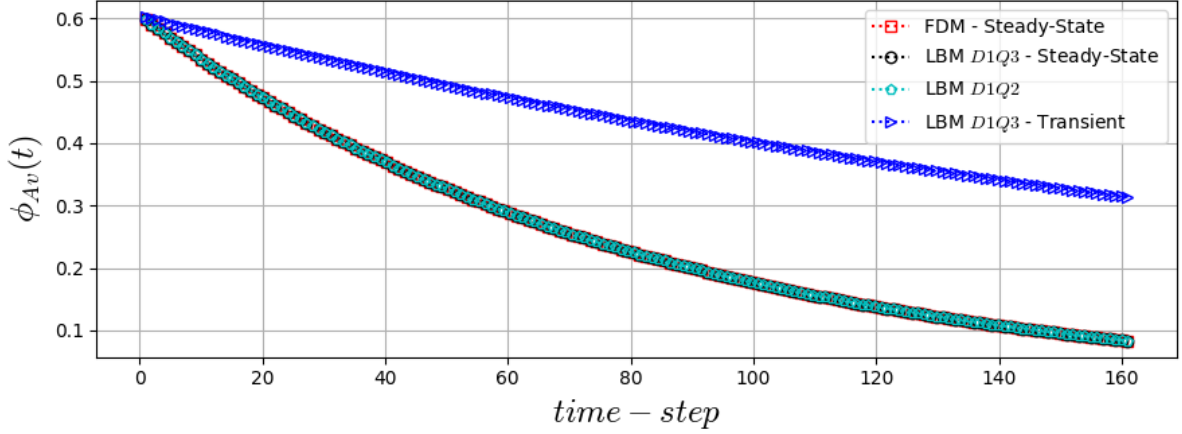
2.2.1 FDM Solution:

2.2.2 LBM D1Q3 Solution With Steady-Scheme:

2.2.3 LBM D1Q2 Solution (Not Accept the Steady-Scheme):

2.2.4 LBM D1Q3 Solution (Transient):

2.2.5 Numerical and Analytical Comparisson:



2.3 2nd Benchmark: 2D Poisson Pressure Field

Analyzing the accuracy of Poisson equation solver, we apply it to a two-dimensional analytical case. The geometry is described by a square domain of length L initialized with a constant dimensionless pressure $\hat{p}(x, y, t = 0) = 0$. The boundary conditions of the geometry are given by

$$\hat{p}(x = 0, y, t) = \frac{\sinh(\pi(1 - y))}{\sinh(\pi)}, \quad \hat{p}(x = L, y, t) = -\frac{\sinh(\pi(1 - y))}{\sinh(\pi)},$$

$$\hat{p}(x, y = 0, t) = \cos(\pi x), \quad \hat{p}(x, y = L, t) = 0.$$

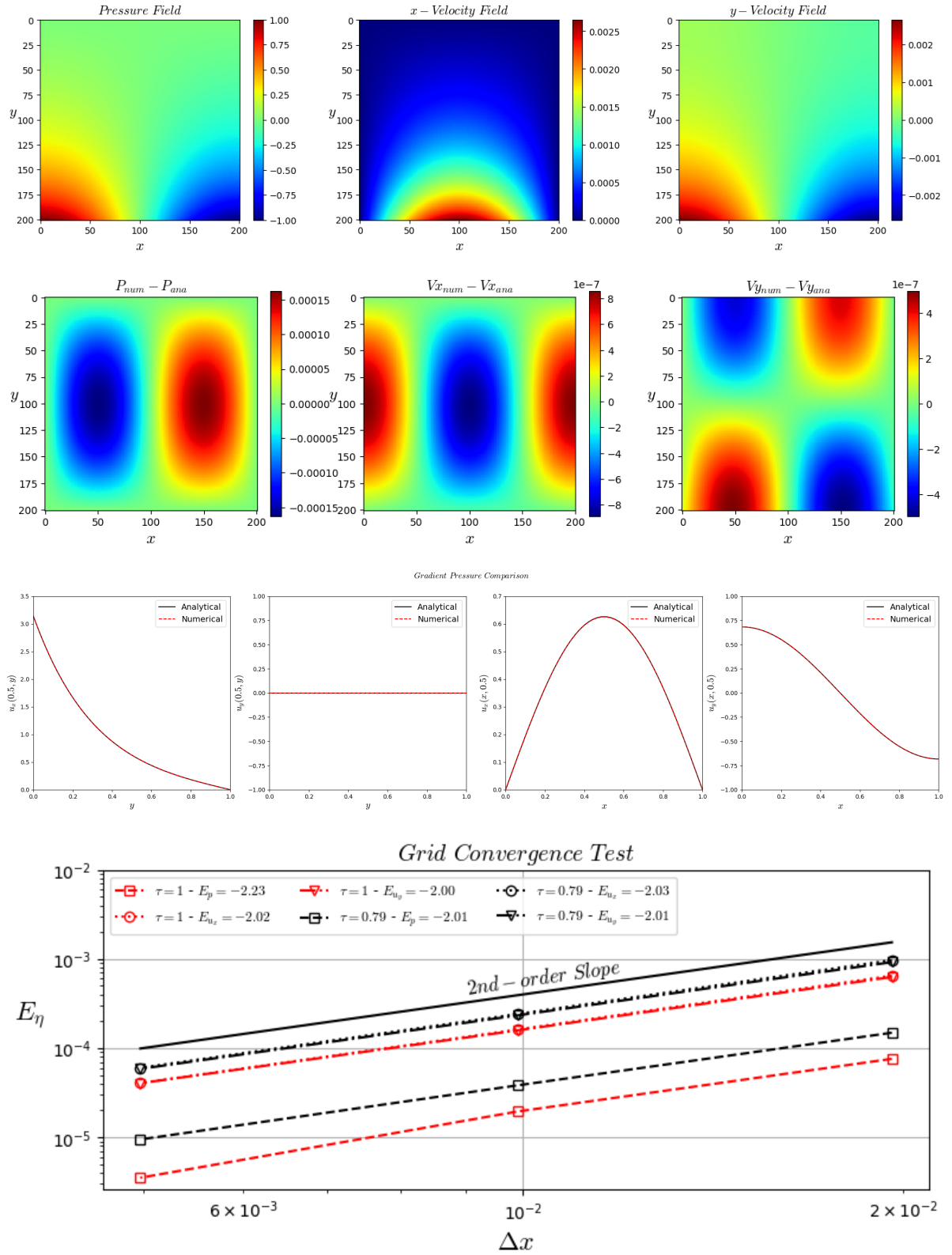
The pressure analytical solution for Eq. (2.1) pressure is given

$$\hat{p}(x, y, t) = \cos(\pi x) \frac{\sinh(\pi(1 - y))}{\sinh(\pi)}$$

where Dp_0 is considered equal to 1, consequently, velocity values can be analytically obtained by the equation:

$$u_x = D\partial_x p = \pi \sin(\pi x) \frac{\sinh(\pi(1 - y))}{\sinh(\pi)}, \quad u_y = D\partial_y p = \pi \cos(\pi x) \frac{\cosh(\pi(1 - y))}{\sinh(\pi)}.$$

2.3.1 LBM D2Q5 Code



CONVECTIVE EQUATION (TRANSPORT EQUATION)

The convective equation also known as transport equation is a partial differential equation (PDE) usually described by

$$\partial_t u + \partial_\alpha B_\alpha = 0 \quad (3.1)$$

where $u(\alpha, t)$ is a time and space dependent parameter that is being transported and $B_\alpha(\alpha, t)$ is a second time and space dependent parameter that can also depend on $u(\alpha, t)$ in non-linear convective diffusive equations.

3.1 Numerical Discretization

Discretizing the above equation in regular grid of 1D domain, we have

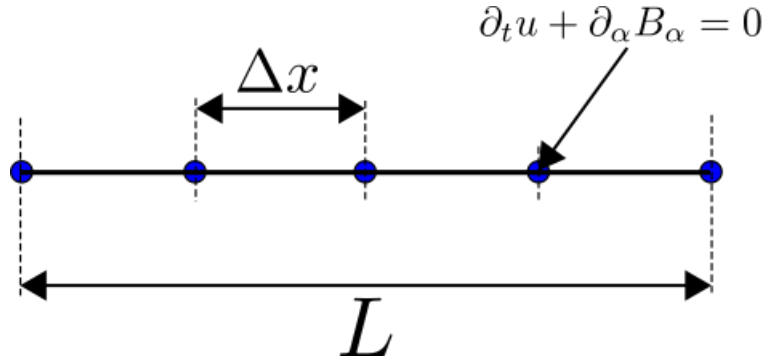


Fig. 3.1: 1D regular grid.

3.1.1 FDM Discretization

The Discretization through finite difference method can assume distinct forms that describe the problem with different accuracy, rate convergence and stability. In the sequence of this section, we will explore different variations that can work with the convective formulation.

Forward-Time and Forward-Space (FTFS)

Discretizing the Eq. (3.1) through finite difference method, we have for forward-time and forward-space (FTFS) discretizations:

$$\partial_t u + \partial_x B_\alpha = \frac{u(x, t + \Delta_t) - u(x, t)}{\Delta_t} + \frac{B_\alpha(x, t) - B_\alpha(x + \Delta_x, t)}{\Delta_x} = 0,$$

isolating the term $u(x, t + \Delta_t)$, we can describe the u time evolution:

$$u(x, t + \Delta_t) = u(x, t) - \frac{\Delta_t}{\Delta_x} (B_\alpha(x, t) - B_\alpha(x + \Delta_x, t)).$$

Forward-Time and Central-Space with Small Diffusivity (FTCS-SD)

Give that discretize convective equation with pure forward-time and central-space (FTCS) can be unstable for several cases, employ a forward-time and central-space with small diffusivity (FTCS-SD) is a alternative to improve stability and ensure a propertie conservation.

$$\begin{aligned} \partial_t u + \partial_x B_\alpha - \partial_\alpha (\nu \partial_\alpha u) = & \frac{u(x, t + \Delta_t) - u(x, t)}{\Delta_t} + \frac{B_\alpha(x + \Delta_x, t) - B_\alpha(x - \Delta_x, t)}{2\Delta_x} \\ & - \nu \frac{u(x - \Delta_x, t) - 2u(x, t) + u(x + \Delta_x, t)}{\Delta_x^2} = 0, \end{aligned}$$

isolating the term $u(x, t + \Delta_t)$, we can describe the u time evolution:

$$u(x, t + \Delta_t) = u(x, t) + \frac{\Delta_t}{2\Delta_x} (B_\alpha(x - \Delta_x, t) - B_\alpha(x + \Delta_x, t)) + \frac{\nu \Delta_t}{\Delta_x^2} (u(x - \Delta_x, t) - 2u(x, t) + u(x + \Delta_x, t)).$$

3.1.2 Lattice Boltzmann Equation

Describing the problem through the BGK lattice Boltzmann equation, we assume a equilibrium state for the present problem, where $f_i = f_i^{eq}$ and the BGK lattice Boltzmann equation reduces to:

$$f_i(x_\alpha + c_{i,\alpha} \Delta t, t + \Delta t) - f_i(x_\alpha, t) = - \left(\frac{f_i - f_i^{eq}}{\tau} \right) \rightarrow f_i(x_\alpha + c_{i,\alpha} \delta t, t + \delta t) = f_i^{eq}, \quad (3.2)$$

where the equilibrium distribution function is defined by

$$f_i^{eq} = w_i \left(u + \frac{c_{i,\alpha} B_\alpha}{c_s^2} \right).$$

Lattice Direction Moments

Raw Moments

$$\begin{array}{llll} \underbrace{\sum_{i=0} f_i^{eq} = \sum_{i=0} f_i = u}_{\text{Zero-Order Moment}} & \underbrace{\sum_{i=0} f_i^{eq} c_{i,x} = B_\alpha}_{\text{x-First-Order Moment}} & \underbrace{\sum_{i=0} f_i^{eq} c_{i,x} c_{i,x} = \frac{u}{3}}_{\text{xx-Second-Order Moment}} & \underbrace{\sum_{i=0} f_i^{eq} c_{i,x} c_{i,x} c_{i,x} = B_\alpha}_{\text{xxx-Third-Order Moment}} \end{array}$$

Hermite Moments

$$\begin{aligned}
 \underbrace{\sum_{i=0} f_i^{eq} = \sum_{i=0} f_i}_{\text{Zero-Order Hermite Moment}} &= u, & \underbrace{\sum_{i=0} f_i^{eq} c_{i,x}}_{\text{x-First-Order Hermite Moment}} &= B_\alpha \\
 \underbrace{\sum_{i=0} f_i^{eq} (c_{i,x} c_{i,x} - c_s^2)}_{\text{xx-Second-Order Hermite Moment}} &= 0, & \underbrace{\sum_{i=0} f_i^{eq} (c_{i,x} c_{i,x} - 3c_s^2) c_{i,x}}_{\text{xxx-Third-Order Hermite Moment}} &= 0
 \end{aligned}$$

Chapmann-Enskog Analysis (Considering Equilibrium State - Consequently Disregarding Inherited Diffusive Effects)

Applying the Chapmann-Enskog procedure to LB equation, we expand the term $f_i(x + e_i \Delta t, t + \Delta t) = f_i'$ in a Taylor series to recover the derivative form of the equation, i.e.,

$$f_i' - f_i(x, t) = \sum_{j=1}^{\infty} \frac{\Delta t^j}{j!} D_t^j f_i = - \left(\frac{f_i - f_i^{eq}}{\tau} \right) \xrightarrow{\text{Equilibrium State}} f_i' = \sum_{j=0}^{\infty} \frac{\Delta t^j}{j!} D_t^j f_i = f_i^{eq}(x, t). \quad (3.3)$$

Rescaling the dimensionless form of the Eq. (3.3) in terms of the Knudsen number (Kn), we have

$$\sum_{j=0}^{\infty} \frac{Kn^{(j-1)}}{j!} D_t^j f_i = f_i^{eq}(x_\alpha, t),$$

applying the asymptotic expansion in both the distribution function ($f_i = f_i^{(0)} + Kn f_i^{(1)} + Kn^2 f_i^{(2)} + \dots$) and time partial derivative ($\partial_t = \partial_t^{(0)} + Kn \partial_t^{(1)} + Kn^2 \partial_t^{(2)} + \dots$), and separating the equation in orders up to the order Kn^2 :

$$\begin{aligned}
 (Kn^{(0)}) : f_i^{(0)} &= f_i^{eq}, \\
 (Kn^{(1)}) : (\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha) f_i^{(0)} &= 0, \\
 (Kn^{(2)}) : \partial_t^{(1)} f_i^{(0)} + (\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha) f_i^{(1)} + \frac{(\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha)^2 f_i^{(0)}}{2} &= 0, \\
 \text{ou} \\
 (Kn^{(2)}) : \partial_t^{(1)} f_i^{(0)} + (\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha) f_i^{(1)} &= 0, \\
 (Kn^{(3)}) : \partial_t^{(2)} f_i^{(0)} + \partial_t^{(1)} f_i^{(1)} + \partial_t^{(1)} ((\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha) f_i^{(0)}) + (\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha) f_i^{(2)} + \frac{(\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha)^2 f_i^{(1)}}{2} \\
 + \frac{(\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha)^3 f_i^{(0)}}{6} &= 0, \\
 \text{ou} \\
 (Kn^{(3)}) : \partial_t^{(2)} f_i^{(0)} + \partial_t^{(1)} f_i^{(1)} + (\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha) f_i^{(2)} &= 0.
 \end{aligned} \quad (3.4)$$

Zero-Order Moment Balance

To retrieve the balance equation, we sum the Eq. (3.4) for $Kn^{(1)}$ and $Kn^{(2)}$ over $\sum_{i=0}$:

$$\begin{aligned}
 (Kn^{(1)}) : \sum (\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha) f_i^{(0)} &= 0, \\
 (Kn^{(1)}) : \partial_t^{(0)} u + \partial_\alpha B_\alpha &= 0,
 \end{aligned} \quad (3.5)$$

$$\begin{aligned}
 (Kn^{(2)}) : \sum [\partial_t^{(1)} f_i^{(0)} + (\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha) f_i^{(1)}] &= 0, \\
 (Kn^{(2)}) : \partial_t^{(1)} u + \partial_\alpha m_\alpha^{(1)} &= 0,
 \end{aligned} \quad (3.6)$$

and

$$\begin{aligned} (Kn^{(3)}) : \quad & \sum_i \left[\partial_t^{(2)} f_i^{(0)} + \partial_t^{(1)} f_i^{(1)} + (\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha) f_i^{(2)} \right] = 0, \\ (Kn^{(3)}) : \quad & \partial_t^{(2)} u + \partial_\alpha m_\alpha^{(2)} = 0. \end{aligned} \quad (3.7)$$

By summing the Eqs. (3.5), (3.6), and (3.7), and also extending to higher orders of Knudsen, we have

$$\partial_t u + \partial_\alpha B_\alpha + \underbrace{\partial_\alpha (m_\alpha^{(1)} + m_\alpha^{(2)} + \dots)}_{\text{Numerical Errors}} = 0,$$

where these numerical error can be related to the numerical error from the approximation of a derivative term by an expansion in Taylor series.

Chapmann-Enskog Analysis (Full)

Applying the Chapmann-Enskog procedure to LB equation, we expand the term $f_i(x + e_i \Delta t, t + \Delta t)$ in a Taylor series to recover the derivative form of the equation, i.e.,

$$f_i(x_\alpha + c_{i,\alpha} \Delta t, t + \Delta t) - f_i(x_\alpha, t) = \sum_{j=1}^{\infty} \frac{\Delta t^j}{j!} D_t^j f_i = - \left(\frac{f_i - f_i^{eq}}{\tau} \right). \quad (3.8)$$

Rescaling the dimensionless form of the Eq. (3.8) in terms of the Knudsen number (Kn), we have

$$\sum_{j=1}^{\infty} \frac{Kn^{(j-1)}}{j!} D_t^j f_i = - \frac{1}{Kn} \left(\frac{f_i - f_{eq,i}}{\tau} \right) \quad \rightarrow \quad \sum_{j=1}^{\infty} \frac{Kn^{(j)}}{j!} D_t^j f_i = - \frac{f_i - f_i^{eq}}{\tau},$$

applying the asymptotic expansion in both the distribution function ($f_i = f_i^{(0)} + Kn f_i^{(1)} + Kn^2 f_i^{(2)} + \dots$) and time partial derivative ($\partial_t = \partial_t^{(0)} + Kn \partial_t^{(1)} + Kn^2 \partial_t^{(2)} + \dots$), and separating the equation in orders up to the order Kn^2 :

$$\begin{aligned} (Kn^{(0)}) : \quad & f_i^{(0)} = f_i^{eq}, \\ (Kn^{(1)}) : \quad & (\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha) f_i^{(0)} = - \frac{f_i^{(1)}}{\tau}, \\ (Kn^{(2)}) : \quad & \partial_t^{(1)} f_i^{(0)} + (\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha) f_i^{(1)} + \frac{(\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha)^2 f_i^{(0)}}{2} = - \frac{f_i^{(2)}}{\tau}, \\ \text{ou} \\ (Kn^{(2)}) : \quad & \partial_t^{(1)} f_i^{(0)} + (\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha) \left(1 - \frac{1}{2\tau} \right) f_i^{(1)} = - \frac{f_i^{(2)}}{\tau} \quad \rightarrow \quad f_i^{(1)} \text{ Formulation} \\ \text{ou} \\ (Kn^{(2)}) : \quad & \partial_t^{(1)} f_i^{(0)} + (\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha)^2 \left(\frac{1}{2} - \tau \right) f_i^{(0)} = - \frac{f_i^{(2)}}{\tau} \quad \rightarrow \quad f_i^{(0)} \text{ Formulation} \\ (Kn^{(3)}) : \quad & \partial_t^{(2)} f_i^{(0)} + \partial_t^{(1)} f_i^{(1)} + \partial_t^{(1)} \left((\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha) f_i^{(0)} \right) + (\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha) f_i^{(2)} + \frac{(\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha)^2 f_i^{(1)}}{2} \\ & + \frac{(\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha)^3 f_i^{(0)}}{6} = - \frac{f_i^{(3)}}{\tau}, \\ \text{ou} \\ (Kn^{(3)}) : \quad & \partial_t^{(2)} f_i^{(0)} + \left(1 - \frac{1}{\tau} \right) \partial_t^{(1)} f_i^{(1)} + \left(\frac{1}{2} - \frac{1}{6\tau} \right) (\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha)^2 f_i^{(1)} + (\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha) f_i^{(2)} = - \frac{f_i^{(3)}}{\tau}, \end{aligned} \quad (3.9)$$

Zero-Order Moment Balance

To retrieve the balance equation, we sum the Eq. (3.9) for $Kn^{(1)}$ and $Kn^{(2)}$ over $\sum_{i=0}$:

$$\begin{aligned} (Kn^{(1)}) : \sum_i (\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha) f_i^{(0)} &= - \sum_i \left(\frac{f_i^{(1)}}{\tau} \right), \\ (Kn^{(1)}) : \partial_t^{(0)} u + \partial_\alpha B_\alpha &= 0, \end{aligned} \quad (3.10)$$

$$\begin{aligned} (Kn^{(2)}) : \sum_i \left[\partial_t^{(1)} f_i^{(0)} + \left(1 - \frac{1}{2\tau} \right) (\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha) f_i^{(1)} \right] &= - \sum_i \left(\frac{f_i^{(2)}}{\tau} \right), \\ (Kn^{(2)}) : \partial_t^{(1)} u + \left(1 - \frac{1}{2\tau} \right) \partial_\alpha m_\alpha^{(1)} &= 0, \end{aligned} \quad (3.11)$$

To determine the unknow term $m_\alpha^{(1)}$, we sum the Eq. (3.9) for $Kn^{(1)}$ over the first-order moment:

$$\begin{aligned} (Kn^{(1)}) : \sum_i (\partial_t^{(0)} + c_{i,\beta} \partial_\beta) f_i^{(0)} c_{i,\alpha} &= - \sum_i \left(\frac{f_i^{(1)}}{\tau} c_{i,\alpha} \right), \\ (Kn^{(1)}) : \partial_t^{(0)} B_\alpha + \partial_\beta (u c_s^2) \delta_{\alpha\beta} &= - \frac{m_\alpha^{(1)}}{\tau} \quad \rightarrow \quad m_\alpha^{(1)} = -\tau (\partial_t^{(0)} B_\alpha + \partial_\beta (u c_s^2) \delta_{\alpha\beta}). \end{aligned} \quad (3.12)$$

Extending the analysis up to $Kn^{(3)}$

$$\begin{aligned} (Kn^{(3)}) : \sum_i \left[\partial_t^{(2)} f_i^{(0)} + \left(1 - \frac{1}{\tau} \right) \partial_t^{(1)} f_i^{(1)} + \left(\frac{1}{2} - \frac{1}{6\tau} \right) (\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha)^2 f_i^{(1)} \right. \\ \left. + (\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha) f_i^{(2)} \right] &= - \sum_i \left[\frac{f_i^{(3)}}{\tau} \right], \\ (Kn^{(3)}) : \partial_t^{(2)} u + \partial_t^{(1)} (\partial_\alpha m_\alpha^{(1)} + \partial_\beta m_\beta^{(1)}) + \partial_\alpha \partial_\beta m_{\alpha\beta}^{(1)} + \partial_\alpha m_\alpha^{(2)} &= 0. \end{aligned} \quad (3.13)$$

By summing the Eqs. (3.10), (3.11), and (3.13), and also extending to higher orders of Knudsen, we have

$$\partial_t u + \partial_\alpha B_\alpha - \partial_\alpha (\nu \partial_\alpha u) + \underbrace{\partial_\alpha m_\alpha^{(2)} + \partial_\alpha \partial_\beta m_{\alpha\beta}^{(1)} + \partial_t^{(1)} (\partial_\alpha m_\alpha^{(1)} + \partial_\beta m_\beta^{(1)}) - \left(\tau - \frac{1}{2} \right) \partial_t^{(0)} \partial_\alpha B_\alpha + \dots}_{\text{Numerical Errors}} = 0,$$

where $\nu = c_s^2(\tau - 1/2)$. The numerical errors can be corrected using extra moments in the equilibrium distribution function and some ones can neglected due to its relevance decay exponentially with the decrease of Δx and Δt .

Boundary Conditions

The boundary conditions for the lattices can be derived by solving a linear system of known moments.

D1Q3: Boundaries	Layers	
	West	East
Unknown f_i	$f_2 = u_e - f_1 - f_0$	$f_1 = u_w - f_2 - f_0$

3.2 Benchmark - 1D Non-Linear Convective formulation of the Buckley-Leverett Equation

Considering a pure convective formulation of the Buckley-Leverett equation, where gravitational effects and capillary pressure are neglected, we have:

$$\partial_t(s_w) + \partial_x \left[\frac{\langle u \rangle_\alpha}{\varphi} F(s_w) \right] = 0, \quad (3.14)$$

with the water fraction flow given by,

$$F(s_w) = \frac{s_w^2}{s_w^2 + M_\mu(1 - s_w)^2}. \quad (3.15)$$

The simulation parameters are chosen so that the average velocity component $\langle u \rangle_\alpha / \varphi$ equals 1 m/s and the dynamic viscosity ratio M_μ equals 1. We consider a one-dimensional porous medium of length $L_x = 20$ m, initially saturated with oil (see Fig. 3.2). Water is injected at the inlet ($x = 0$) under a Dirichlet boundary condition $s_w(x = 0) = 1$. At the outlet ($x = L$), a Neumann boundary condition specifying a zero gradient in water saturation is applied.

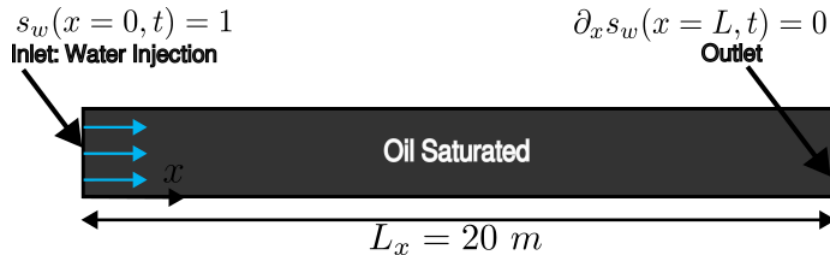


Fig. 3.2: Geometry of the non-linear convective problem: initial and boundary conditions.

3.2.1 Transient Analytical Solution

The non-linear convective problem given by the Eq. (3.14) has an analytical solution dependent of the initial condition (Riemann problem). In the present case, the solution for the specific initial condition is given by

$$s_w(x, t) = \begin{cases} tF' & \text{for } x \leq \zeta t \\ 0 & \text{for } x > \zeta t \end{cases}$$

where $\zeta = F'(\tilde{s}_w) = F(\tilde{s}_w)/\tilde{s}_w$ is the shock wave speed, and $\tilde{s}_w$ is the sharp front saturation determined analytically by

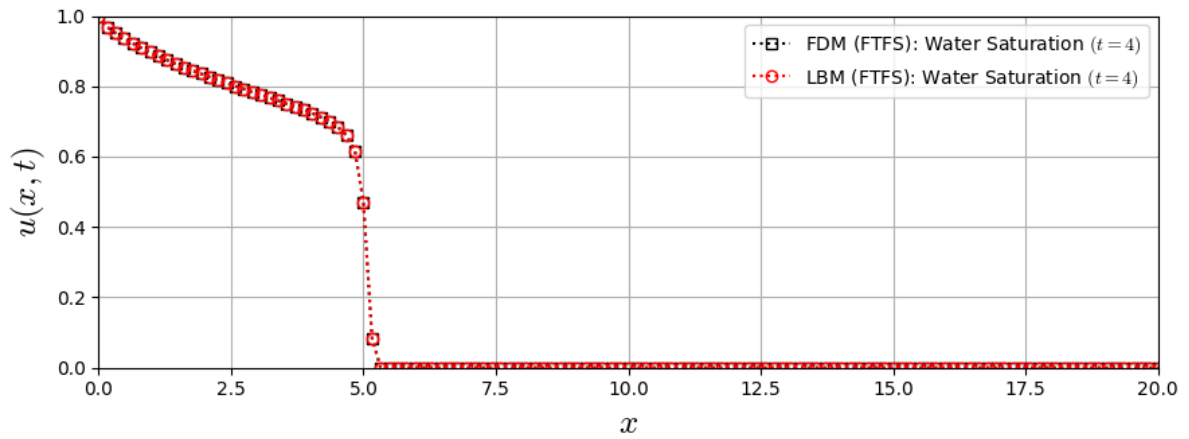
$$\tilde{s}_w = \frac{F(\tilde{s}_w)}{F'(\tilde{s}_w)} = \sqrt{\frac{M_\mu}{M_\mu + 1}}, \quad \text{where} \quad F'(s_w) = \frac{dF}{ds_w} = \frac{2s_w M_\mu (1 - s_w)}{(s_w^2 + M_\mu(1 - s_w)^2)^2}.$$

Forward Time and Forward Space - FTFS

FDM Solution:

Lattice Boltzmann Framework of FTFS:

Numerical and Analytical Comparisson

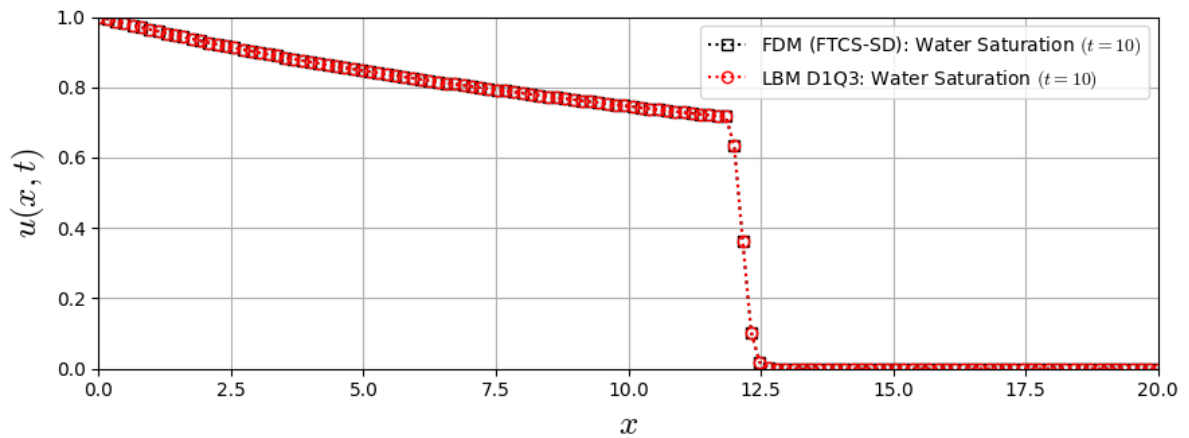


3.2.2 Forward-Time and Central-Space with Small Diffusivity (FTCS-SD)

Lattice Boltzmann Method (D1Q3):

FDM approach of LBM Framework D1Q3:

Numerical and Analytical Comparisson



CONVECTIVE-DIFFUSIVE EQUATION

The convective-diffusive equation is a partial differential equation (PDE) commonly written as

$$\partial_t u + \partial_\alpha B_\alpha - \partial_\alpha (\nu \partial_\alpha C) = 0 \quad (4.1)$$

where $u(\alpha, t)$ is a space and time dependent scalar parameter, $B_\alpha(\alpha, t)$ is a vector a C a scalar term, may be time and space dependent parameters and can also be also expressed as linear or non-linear functions of $u(\alpha, t)$. The term ν is the diffusion coefficient, a constitutive term that controls the intensity of diffusive effect.

4.1 Numerical Discretization

Discretizing the above equation in regular grid of 1D domain, we have

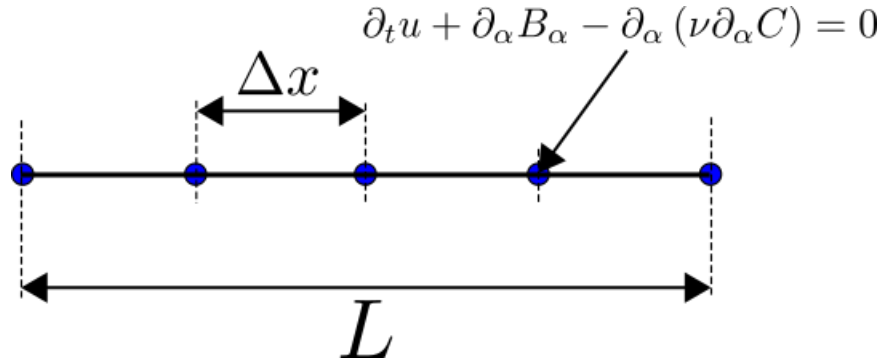


Fig. 4.1: 1D regular grid.

4.1.1 FDM Discretization

To explore different discretization approaches, we adopt forward-time, forward-space (commonly referred to as the up-wind scheme), as well as central-difference schemes.

Forward-Time and Forward-Space (FTFSCS)

Discretizing the Eq. (4.1) through finite difference method, we have for forward-time, forward-space for advection and central-space for diffusion (FTFSCS) discretizations:

$$\partial_t u + \partial_\alpha B_\alpha - \partial_\alpha (\nu \partial_\alpha C) = \frac{u(x, t + \Delta_t) - u(x, t)}{\Delta_t} + \frac{B_\alpha(x, t) - B_\alpha(x + \Delta_x, t)}{\Delta_x} - \nu \frac{C(x - \Delta_x, t) - 2C(x, t) + C(x + \Delta_x, t)}{\Delta_x^2} = 0,$$

isolating the term $u(x, t + \Delta_t)$, we can describe the u time evolution:

$$u(x, t + \Delta_t) = u(x, t) - \frac{\Delta_t}{\Delta_x} (B_\alpha(x, t) - B_\alpha(x + \Delta_x, t)) + \frac{\nu \Delta_t}{\Delta_x^2} (C(x - \Delta_x, t) - 2C(x, t) + C(x + \Delta_x, t))$$

Forward-Time and Central-Space with Small Diffusivity (FTCS)

Given the discretization of the convective-diffusive equation with pure forward-time and central-space in advective and diffusive terms (FTCS), we have

$$\partial_t u + \partial_x B_\alpha - \partial_\alpha (\nu \partial_\alpha u) = \frac{u(x, t + \Delta_t) - u(x, t)}{\Delta_t} + \frac{B_\alpha(x + \Delta_x, t) - B_\alpha(x - \Delta_x, t)}{2\Delta_x} - \nu \frac{C(x - \Delta_x, t) - 2C(x, t) + C(x + \Delta_x, t)}{\Delta_x^2} = 0,$$

isolating the term $u(x, t + \Delta_t)$, we can describe the u time evolution:

$$u(x, t + \Delta_t) = u(x, t) - \frac{\Delta_t}{2\Delta_x} (B_\alpha(x + \Delta_x, t) - B_\alpha(x - \Delta_x, t)) + \frac{\nu \Delta_t}{\Delta_x^2} (C(x - \Delta_x, t) - 2C(x, t) + C(x + \Delta_x, t))$$

Forward-Time and Central-Space with Small Diffusivity (FTCS-SD)

Give that discretize convective equation with pure forward-time and central-space (FTCS), employ a small diffusivity (FTCS-SD) is a alternative to improve stability and ensure a propertie conservation.

$$\partial_t u + \partial_x B_\alpha - \partial_\alpha (\nu_u \partial_\alpha u) - \partial_\alpha (\nu \partial_\alpha C) = \frac{u(x, t + \Delta_t) - u(x, t)}{\Delta_t} + \frac{B_\alpha(x + \Delta_x, t) - B_\alpha(x - \Delta_x, t)}{2\Delta_x} - \nu_u \frac{u(x - \Delta_x, t) - 2u(x, t) + u(x + \Delta_x, t)}{\Delta_x^2} - \nu \frac{C(x - \Delta_x, t) - 2C(x, t) + C(x + \Delta_x, t)}{\Delta_x^2} = 0,$$

isolating the term $u(x, t + \Delta_t)$, we can describe the u time evolution:

$$u(x, t + \Delta_t) = u(x, t) - \frac{\Delta_t}{2\Delta_x} (B_\alpha(x + \Delta_x, t) - B_\alpha(x - \Delta_x, t)) + \frac{\nu_u \Delta_t}{\Delta_x^2} (u(x - \Delta_x, t) - 2u(x, t) + u(x + \Delta_x, t)) + \frac{\nu \Delta_t}{\Delta_x^2} (C(x - \Delta_x, t) - 2C(x, t) + C(x + \Delta_x, t)).$$

4.1.2 Lattice Boltzmann Equation

Describing the problem through the BGK lattice Boltzmann equation:

$$f_i(x_\alpha + c_{i,\alpha} \Delta t, t + \Delta t) - f_i(x_\alpha, t) = - \left(\frac{f_i - f_i^{eq}}{\tau} \right). \quad (4.2)$$

The equilibrium distribution function is defined by

$$f_i^{eq} = w_i \left(u + \frac{c_{i,\alpha} B_\alpha}{c_s^2} + \frac{(c_s^2 C \delta_{\alpha\beta} + \Lambda_{\alpha\beta} - c_s^2 u \delta_{\alpha\beta}) (c_{i,\alpha} c_{i,\beta} - c_s^2 \delta_{\alpha\beta})}{2c_s^4} \right),$$

where $\Lambda_{\alpha\beta}$ is a correction term defined by

$$\Lambda_{\alpha\beta} = \int B'_\alpha B'_\beta du = \int \left(\frac{\partial B_\alpha}{\partial u} \frac{\partial B_\beta}{\partial u} \right) du.$$

Lattice Direction Moments

Raw Moments

$$\underbrace{\sum_{i=0} f_i^{eq}}_{\text{Zero-Order Moment}} = \sum_{i=0} f_i = u, \quad \underbrace{\sum_{i=0} f_i^{eq} c_{i,x}}_{\text{x-First-Order Moment}} = B_\alpha, \quad \underbrace{\sum_{i=0} f_i^{eq} c_{i,x} c_{i,x}}_{\text{xx-Second-Order Moment}} = \frac{C}{3} + \Lambda_{\alpha\beta}, \quad \underbrace{\sum_{i=0} f_i^{eq} c_{i,x} c_{i,x} c_{i,x}}_{\text{xxx-Third-Order Moment}} = B_\alpha.$$

Hermite Moments

$$\underbrace{\sum_{i=0} f_i^{eq}}_{\text{Zero-Order Hermite Moment}} = \sum_{i=0} f_i = u, \quad \underbrace{\sum_{i=0} f_i^{eq} c_{i,x}}_{\text{x-First-Order Hermite Moment}} = B_\alpha,$$

$$\underbrace{\sum_{i=0} f_i^{eq} (c_{i,x} c_{i,x} - c_s^2)}_{\text{xx-Second-Order Hermite Moment}} = \frac{C}{3} + \Lambda_{\alpha\beta} - \frac{u}{3}, \quad \underbrace{\sum_{i=0} f_i^{eq} (c_{i,x} c_{i,x} - 3c_s^2) c_{i,x}}_{\text{xxx-Third-Order Hermite Moment}} = 0.$$

Chapmann-Enskog Analysis

Applying the Chapmann-Enskog procedure to LB equation, we expand the term $f_i(x + e_i \Delta t, t + \Delta t)$ in a Taylor series to recover the derivative form of the equation, i.e.,

$$f_i(x_\alpha + c_{i,\alpha} \Delta t, t + \Delta t) - f_i(x_\alpha, t) = \sum_{j=1}^{\infty} \frac{\Delta t^j}{j!} D_t^j f_i = - \left(\frac{f_i - f_i^{eq}}{\tau} \right). \quad (4.3)$$

Rescaling the dimensionless form of the Eq. (4.3) in terms of the Knudsen number (Kn), we have

$$\sum_{j=1}^{\infty} \frac{Kn^{(j-1)}}{j!} D_t^j f_i = - \frac{1}{Kn} \left(\frac{f_i - f_{eq,i}}{\tau} \right) \quad \rightarrow \quad \sum_{j=1}^{\infty} \frac{Kn^{(j)}}{j!} D_t^j f_i = - \frac{f_i - f_i^{eq}}{\tau},$$

applying the asymptotic expansion in both the distribution function ($f_i = f_i^{(0)} + Kn f_i^{(1)} + Kn^2 f_i^{(2)} + \dots$) and time partial derivative ($\partial_t = \partial_t^{(0)} + Kn \partial_t^{(1)} + Kn^2 \partial_t^{(2)} + \dots$), and separating the equation in orders up to the order Kn^2 :

$$\begin{aligned}
 (Kn^{(0)}) : \quad & f_i^{(0)} = f_i^{eq}, \\
 (Kn^{(1)}) : \quad & (\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha) f_i^{(0)} = -\frac{f_i^{(1)}}{\tau}, \\
 (Kn^{(2)}) : \quad & \partial_t^{(1)} f_i^{(0)} + (\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha) f_i^{(1)} + \frac{(\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha)^2 f_i^{(0)}}{2} = -\frac{f_i^{(2)}}{\tau}, \\
 \text{ou} \\
 (Kn^{(2)}) : \quad & \partial_t^{(1)} f_i^{(0)} + (\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha) \left(1 - \frac{1}{2\tau}\right) f_i^{(1)} = -\frac{f_i^{(2)}}{\tau} \quad \rightarrow \quad f_i^{(1)} \text{ Formulation} \\
 \text{ou} \\
 (Kn^{(2)}) : \quad & \partial_t^{(1)} f_i^{(0)} + (\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha)^2 \left(\frac{1}{2} - \tau\right) f_i^{(0)} = -\frac{f_i^{(2)}}{\tau} \quad \rightarrow \quad f_i^{(0)} \text{ Formulation} \\
 (Kn^{(3)}) : \quad & \partial_t^{(2)} f_i^{(0)} + \partial_t^{(1)} f_i^{(1)} + \partial_t^{(1)} \left((\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha) f_i^{(0)}\right) + (\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha) f_i^{(2)} + \frac{(\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha)^2 f_i^{(1)}}{2} \\
 & + \frac{(\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha)^3 f_i^{(0)}}{6} = -\frac{f_i^{(3)}}{\tau}, \\
 \text{ou} \\
 (Kn^{(3)}) : \quad & \partial_t^{(2)} f_i^{(0)} + \left(1 - \frac{1}{\tau}\right) \partial_t^{(1)} f_i^{(1)} + \left(\frac{1}{2} - \frac{1}{6\tau}\right) (\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha)^2 f_i^{(1)} + (\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha) f_i^{(2)} = -\frac{f_i^{(3)}}{\tau}, \\
 & \hspace{15em} (4.4)
 \end{aligned}$$

Zero-Order Moment Balance

To retrieve the balance equation, we sum the Eq. (4.4) for $Kn^{(1)}$ and $Kn^{(2)}$ over $\sum_{i=0}$:

$$\begin{aligned}
 (Kn^{(1)}) : \quad & \sum_i (\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha) f_i^{(0)} = -\sum_i \left(\frac{f_i^{(1)}}{\tau}\right), \\
 (Kn^{(1)}) : \quad & \partial_t^{(0)} u + \partial_\alpha B_\alpha = 0, \\
 (Kn^{(2)}) : \quad & \sum_i \left[\partial_t^{(1)} f_i^{(0)} + \left(1 - \frac{1}{2\tau}\right) (\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha) f_i^{(1)} \right] = -\sum_i \left(\frac{f_i^{(2)}}{\tau}\right), \\
 (Kn^{(2)}) : \quad & \partial_t^{(1)} u + \left(1 - \frac{1}{2\tau}\right) \partial_\alpha m_\alpha^{(1)} = 0,
 \end{aligned}
 \tag{4.5}$$

To determine the unknow term $m_\alpha^{(1)}$, we sum the Eq. (4.4) for $Kn^{(1)}$ over the first-order moment:

$$\begin{aligned}
 (Kn^{(1)}) : \quad & \sum_i (\partial_t^{(0)} + c_{i,\beta} \partial_\beta) f_i^{(0)} c_{i,\alpha} = -\sum_i \left(\frac{f_i^{(1)}}{\tau} c_{i,\alpha}\right), \\
 (Kn^{(1)}) : \quad & \partial_t^{(0)} B_\alpha + \partial_\beta (c_s^2 C \delta_{\alpha\beta} + \Lambda_{\alpha\beta}) = -\frac{m_\alpha^{(1)}}{\tau} \quad \rightarrow \quad m_\alpha^{(1)} = -\tau \partial_\beta (c_s^2 C \delta_{\alpha\beta}).
 \end{aligned}
 \tag{4.6}$$

Extending the analysis up to $Kn^{(3)}$

$$\begin{aligned}
 (Kn^{(3)}) : \quad & \sum_i \left[\partial_t^{(2)} f_i^{(0)} + \left(1 - \frac{1}{\tau}\right) \partial_t^{(1)} f_i^{(1)} + \left(\frac{1}{2} - \frac{1}{6\tau}\right) (\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha)^2 f_i^{(1)} \right. \\
 & \left. + (\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha) f_i^{(2)} \right] = -\sum_i \left(\frac{f_i^{(3)}}{\tau}\right), \\
 (Kn^{(3)}) : \quad & \partial_t^{(2)} u + \partial_t^{(1)} (\partial_\alpha m_\alpha^{(1)} + \partial_\beta m_\beta^{(1)}) + \partial_\alpha \partial_\beta m_{\alpha\beta}^{(1)} + \partial_\alpha m_\alpha^{(2)} = 0.
 \end{aligned}
 \tag{4.7}$$

By summing the Eqs. (4.5), (4.6), and (4.7), and also extending to higher orders of Knudsen, we have

$$\partial_t u + \partial_\alpha B_\alpha - \partial_\alpha (\nu \partial_\alpha C) + \underbrace{\partial_\alpha m_\alpha^{(2)} + \partial_\alpha \partial_\beta m_{\alpha\beta}^{(1)} + \partial_t^{(1)} (\partial_\alpha m_\alpha^{(1)} + \partial_\beta m_\beta^{(1)})}_{\text{Numerical Errors}} + \dots = 0,$$

where $\nu = c_s^2(\tau - 1/2)$. The numerical errors can be corrected using extra moments in the equilibrium distribution function and some ones can neglected due to its relevance decay exponentially with the decrease of Δx and Δt .

Boundary Conditions

The boundary conditions for the lattices can be derived by solving a linear system of known moments.

D1Q3: Boundaries	Layers	
	West	East
Unknown f_i	$f_2 = u_e - f_1 - f_0$	$f_1 = u_w - f_2 - f_0$

4.2 1st Benchmark: Decay of Sinoidal Wave

Considering a 1D **convection–diffusion equation** with constant coefficients $\frac{\partial u}{\partial t} + c \frac{\partial u}{\partial x} = \nu \frac{\partial^2 u}{\partial x^2}$, $x \in \mathbb{R}$, $t \geq 0$, with **sinusoidal initial condition**

$$u(x, 0) = A \sin(kx + \phi),$$

where A is the initial amplitude, k the wavenumber, ϕ a phase shift.

Then the analytical solution is

$$u(x, t) = Ae^{-\nu k^2 t} \sin(k(x - ct) + \phi)$$

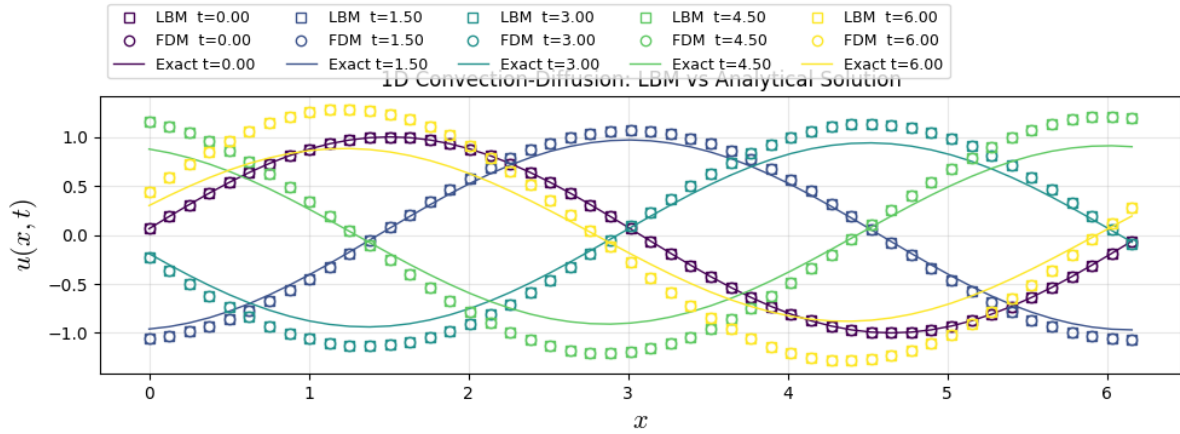
So:

- **Convection** shifts the wave: $x \mapsto x - ct$ (wave travels at speed c).
- **Diffusion** damps the amplitude exponentially: $A(t) = Ae^{-\nu k^2 t}$.

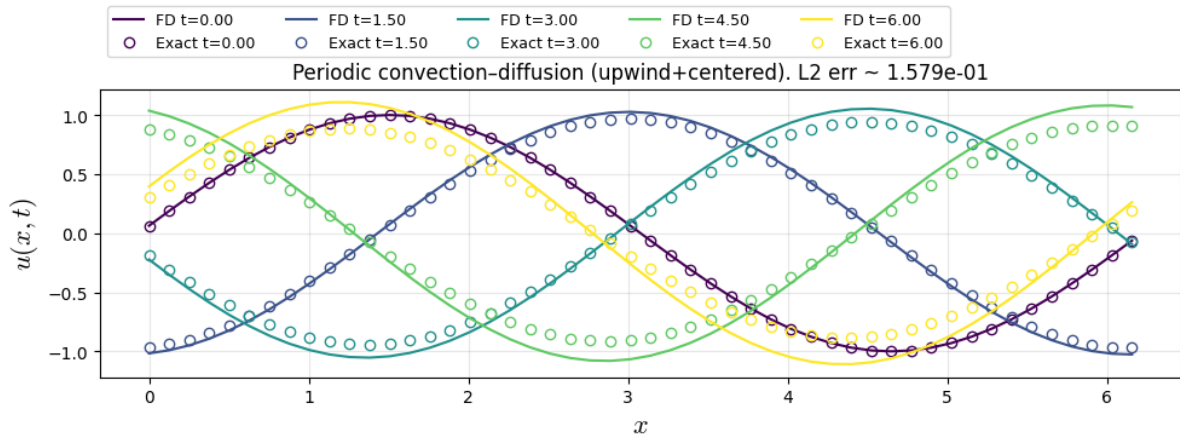
If you use a cosine initial condition $u(x, 0) = A \cos(kx + \phi)$, the solution is the same form: $u(x, t) = Ae^{-\nu k^2 t} \cos(k(x - ct) + \phi)$.

4.2.1 LBM D1Q3 Solution

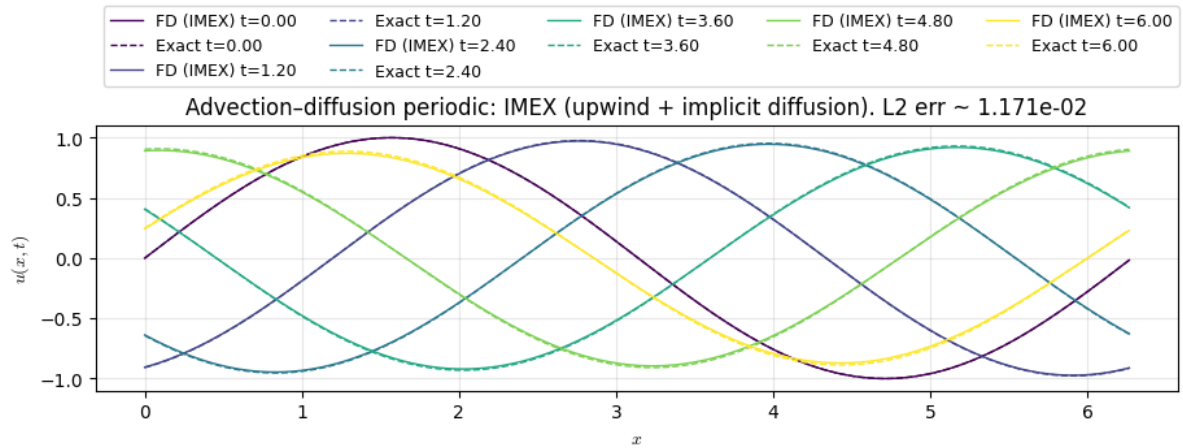
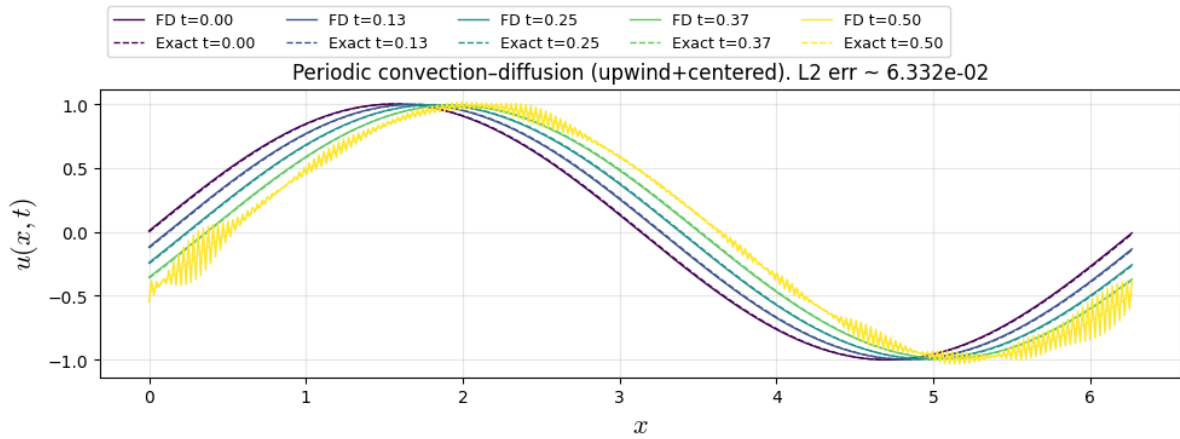
4.2.2 FDM Solution (FTCS-SD) (LBM Framework $\tau = 1$)



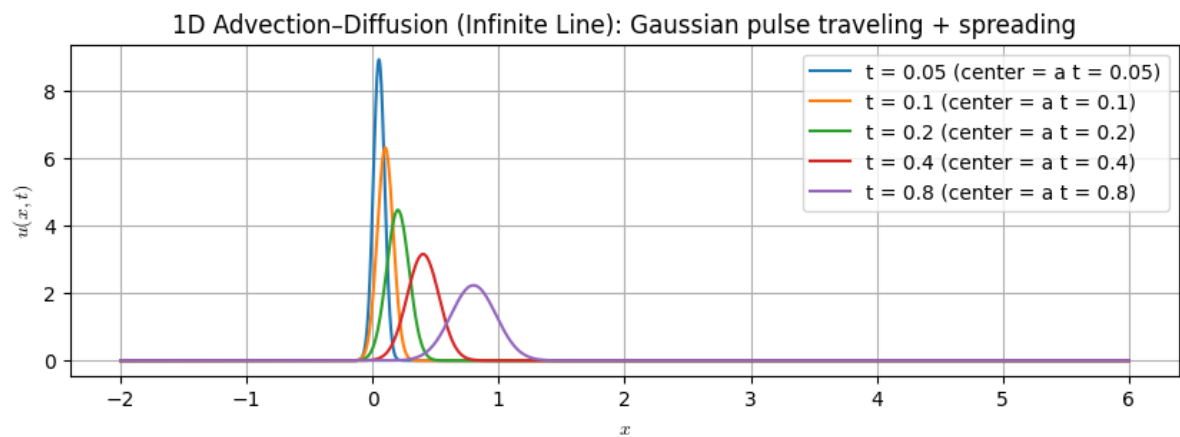
4.2.3 FDM Solution (FTCS)

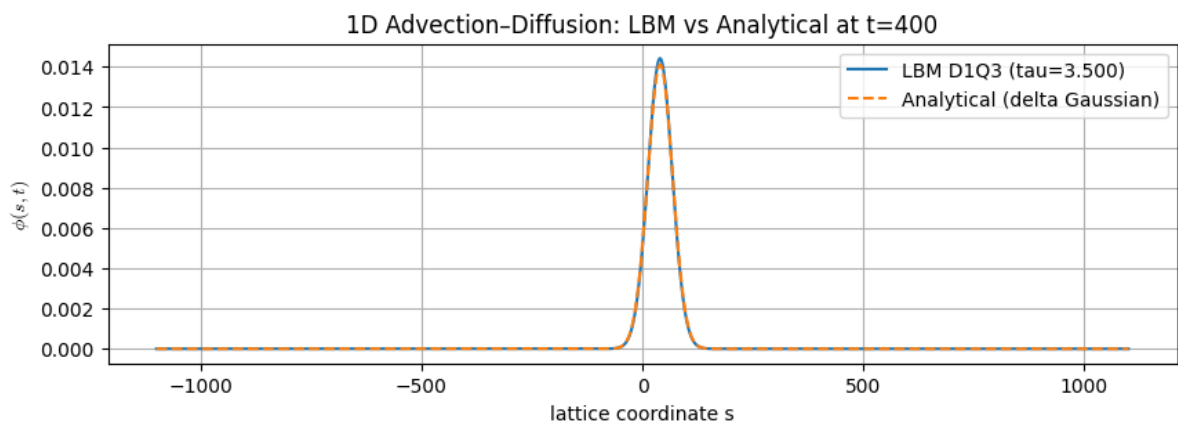
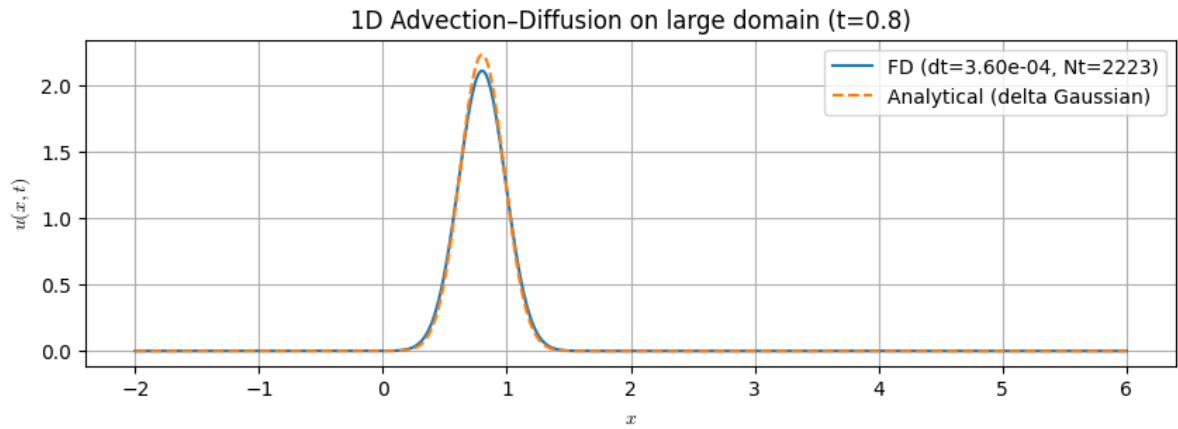


4.2.4 FDM Solution (FTFSCS)



4.3 2nd Benchmark: 1D Perturbation wave-decay





NAVIER-STOKES EQUATION

The Navier-Stokes equation is a non-linear PDE able to describe mass and momentum balance of fluid flow problems, commonly written as

$$\underbrace{\partial_t \rho + \partial_\alpha \rho u_\alpha}_{\text{Mass Balance Equation}} = 0,$$

$$\underbrace{\partial_t \rho u_\alpha + \partial_\beta (\rho u_\alpha u_\beta)}_{\text{Momentum Balance Equation}} = \partial_\beta (-p \delta_{\alpha\beta} + \mu (\partial_\alpha u_\beta + \partial_\beta u_\alpha)) + \rho g_\alpha, \quad (5.1)$$

where ρ is the fluid density, u_α is the α -th component of the velocity vector field, p is the pressure field, $\delta_{\alpha\beta}$ is the Kronecker delta, μ is the dynamic viscosity (a constitutive parameter controlling the intensity of viscous diffusion), and g_α is the α -th component of the external body acceleration (e.g., gravity).

5.1 Numerical Discretization: Lattice Boltzmann Equation

Describing the problem through the BGK lattice Boltzmann equation:

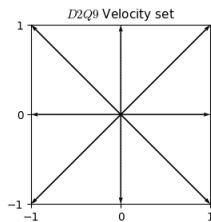
$$\underbrace{f_i(x_\alpha + c_{i,\alpha} \Delta t, t + \Delta t) - f_i(x_\alpha, t)}_{\text{1st order space-time discretization}} = - \left(\frac{f_i - f_i^{eq}}{\tau} \right). \quad (5.2)$$

$$\underbrace{\hat{f}_i(x_\alpha + c_{i,\alpha} \Delta t, t + \Delta t) - \hat{f}_i(x_\alpha, t)}_{\text{2nd order space-time discretization}} = - \left(\frac{\hat{f}_i - \hat{f}_i^{eq}}{\tau} \right) + \left(1 - \frac{1}{2\tau} F_i \right). \quad (5.3)$$

The equilibrium distribution function is defined by

$$f_i^{eq} = w_i \rho \left(1 + \frac{c_{i,\alpha} u_\alpha}{c_s^2} + \frac{u_\alpha u_\beta (c_{i,\alpha} c_{i,\beta} - c_s^2 \delta_{\alpha\beta})}{2c_s^4} \right).$$

5.1.1 Lattice Direction Moments: D2Q9 Velocity-Space Discetization



D2Q9 lattice properties

i	c_x	c_y	Directions	$ c $	w_i (exact)	w_i (float)
0	0	0	rest	0 (rest)	$\frac{1}{9}$	0.444444
1	1	0	East	1 (face)	$\frac{1}{6}$	0.111111
2	-1	0	West	1 (face)	$\frac{1}{6}$	0.111111
3	0	1	North	1 (face)	$\frac{1}{6}$	0.111111
4	0	-1	South	1 (face)	$\frac{1}{6}$	0.111111
5	1	1	NE	$\sqrt{2}$ (Diag)	$\frac{1}{18}$	0.027778
6	-1	1	NW	$\sqrt{2}$ (Diag)	$\frac{1}{18}$	0.027778
7	-1	-1	SW	$\sqrt{2}$ (Diag)	$\frac{1}{18}$	0.027778
8	1	-1	SE	$\sqrt{2}$ (Diag)	$\frac{1}{18}$	0.027778

Hermite hierarchy

$\mathcal{H}^0$			
$\mathcal{H}_x$	$\mathcal{H}_y$		
$\mathcal{H}_{xx}$	$\mathcal{H}_{xy}$	$\mathcal{H}_{yy}$	
$\mathcal{H}_{xxy}$	$\mathcal{H}_{xyy}$		
$\mathcal{H}_{xxyy}$			

Raw Moments

$$\begin{aligned}
 \underbrace{\sum_{i=0} f_i^{eq}}_{\text{Zero-Order Moment}} &= \sum_{i=0} f_i = \rho, & \underbrace{\sum_{i=0} f_i^{eq} c_{i,x}}_{\text{x-First-Order Moment}} &= \rho u_x, & \underbrace{\sum_{i=0} f_i^{eq} c_{i,y}}_{\text{y-First-Order Moment}} &= \rho u_y \\
 \underbrace{\sum_{i=0} f_i^{eq} c_{i,x} c_{i,x}}_{\text{xx-Second-Order Moment}} &= \rho \left(u_x^2 + \frac{1}{3} \right), & \underbrace{\sum_{i=0} f_i^{eq} c_{i,x} c_{i,y}}_{\text{xy-Second-Order Moment}} &= \rho u_x u_y, & \underbrace{\sum_{i=0} f_i^{eq} c_{i,y} c_{i,y}}_{\text{yy-Second-Order Moment}} &= \rho \left(u_y^2 + \frac{1}{3} \right) \\
 \underbrace{\sum_{i=0} f_i^{eq} c_{i,x} c_{i,x} c_{i,x}}_{\text{xxx-Third-Order Moment}} &= \rho u_x, & \underbrace{\sum_{i=0} f_i^{eq} c_{i,x} c_{i,x} c_{i,y}}_{\text{xxy-Third-Order Moment}} &= \frac{\rho u_y}{3}, & \underbrace{\sum_{i=0} f_i^{eq} c_{i,x} c_{i,y} c_{i,y}}_{\text{xyy-Third-Order Moment}} &= \frac{\rho u_x}{3}, & \underbrace{\sum_{i=0} f_i^{eq} c_{i,y} c_{i,y} c_{i,y}}_{\text{yyy-Third-Order Moment}} &= \rho u_y
 \end{aligned}$$

Hermite Moments

$$\begin{aligned}
 \underbrace{\sum_{i=0} f_i^{eq}}_{\text{Zero-Order Hermite Moment}} &= \sum_{i=0} f_i = \rho, & \underbrace{\sum_{i=0} f_i^{eq} c_{i,x}}_{\text{x-First-Order Hermite Moment}} &= \rho u_x, & \underbrace{\sum_{i=0} f_i^{eq} c_{i,y}}_{\text{y-First-Order Hermite Moment}} &= \rho u_y \\
 \underbrace{\sum_{i=0} f_i^{eq} c_{i,x} c_{i,x}}_{\text{xx-Second-Order Moment}} &= \rho u_x^2, & \underbrace{\sum_{i=0} f_i^{eq} c_{i,x} c_{i,y}}_{\text{xy-Second-Order Moment}} &= \rho u_x u_y, & \underbrace{\sum_{i=0} f_i^{eq} c_{i,y} c_{i,y}}_{\text{yy-Second-Order Moment}} &= \rho u_y^2 \\
 \underbrace{\sum_{i=0} f_i^{eq} c_{i,x} c_{i,x} c_{i,x}}_{\text{xxx-Third-Order Moment}} &= 0, & \underbrace{\sum_{i=0} f_i^{eq} c_{i,x} c_{i,x} c_{i,y}}_{\text{xxy-Third-Order Moment}} &= 0, & \underbrace{\sum_{i=0} f_i^{eq} c_{i,x} c_{i,y} c_{i,y}}_{\text{xyy-Third-Order Moment}} &= 0, & \underbrace{\sum_{i=0} f_i^{eq} c_{i,y} c_{i,y} c_{i,y}}_{\text{yyy-Third-Order Moment}} &= 0
 \end{aligned}$$

5.1.2 Chapmann-Enskog Analysis: 1st order Lattice Boltzmann Equation

Applying the Chapmann-Enskog procedure to LB equation, we expand the term $f_i(x + c_{i,\alpha} \Delta t, t + \Delta t)$ in a Taylor series to recover the derivative form of the equation, i.e.,

$$f_i(x_\alpha + c_{i,\alpha} \Delta t, t + \Delta t) - f_i(x_\alpha, t) = \sum_{j=1}^{\infty} \frac{\Delta t^j}{j!} D_t^j f_i = - \left(\frac{f_i - f_i^{eq}}{\tau} \right), \quad (5.4)$$

where $D_t f_i = \partial_t f_i + c_{i,\alpha} \partial_\alpha f_i$.

Rescaling to the dimensionless form of the Eq. (5.4), we rewrite it in terms of the Knudsen number (Kn) following the procedure presented in Kremer [Kre10] Section 3.3.1:

$$\sum_{j=1}^{\infty} \frac{Kn^{(j-1)}}{j!} D_t^j f_i = - \frac{1}{Kn} \left(\frac{f_i - f_{eq,i}}{\tau} \right) \quad \rightarrow \quad \sum_{j=1}^{\infty} \frac{Kn^{(j)}}{j!} D_t^j f_i = - \frac{f_i - f_i^{eq}}{\tau},$$

expanding the above equation up to fourth-order:

$$\begin{aligned}
 \frac{f_i^{eq} - f_i}{\tau} &= Kn^{(1)} (c_{i,\alpha} \partial_\alpha f_i + \partial_t f_i) \\
 &+ Kn^{(2)} \left(\frac{c_{i,\alpha}^2 \partial_\alpha^2 f_i}{2} + c_{i,\alpha} \partial_\alpha \partial_t f_i + \frac{\partial_t^2 f_i}{2} \right) \\
 &+ Kn^{(3)} \left(\frac{c_{i,\alpha}^3 \partial_\alpha^3 f_i}{6} + \frac{c_{i,\alpha}^2 \partial_\alpha^2 \partial_t f_i}{2} + \frac{c_{i,\alpha} \partial_\alpha \partial_t^2 f_i}{2} + \frac{\partial_t^3 f_i}{6} \right) \\
 &+ Kn^{(4)} \left(\frac{c_{i,\alpha}^4 \partial_\alpha^4 f_i}{24} + \frac{c_{i,\alpha}^3 \partial_\alpha^3 \partial_t f_i}{6} + \frac{c_{i,\alpha}^2 \partial_\alpha^2 \partial_t^2 f_i}{4} + \frac{c_{i,\alpha} \partial_\alpha \partial_t^3 f_i}{6} + \frac{\partial_t^4 f_i}{24} \right)
 \end{aligned}$$

Applying the asymptotic expansion in both the distribution function ($f_i = f_i^{(0)} + Kn f_i^{(1)} + Kn^2 f_i^{(2)} + \dots$) and time partial derivative ($\partial_t = \partial_t^{(0)} + Kn \partial_t^{(1)} + Kn^2 \partial_t^{(2)} + \dots$):

$$\begin{aligned}
 &-\frac{f_i^{(0)}}{\tau} + \frac{f_i^{eq}}{\tau} - \frac{Kn f_i^{(1)}}{\tau} - \frac{Kn^2 f_i^{(2)}}{\tau} - \frac{Kn^3 f_i^{(3)}}{\tau} - \frac{Kn^4 f_i^{(4)}}{\tau} = \\
 &Kn (c_{i,\alpha} \partial_\alpha f_i^{(0)} + \partial_t^{(0)} f_i^{(0)}) \\
 &+ Kn^2 \left(\frac{c_{i,\alpha}^2 \partial_\alpha^2 f_i^{(0)}}{2} + c_{i,\alpha} \partial_\alpha f_i^{(1)} + c_{i,\alpha} \partial_\alpha \partial_t^{(0)} f_i^{(0)} + \frac{(\partial_t^{(0)})^2 f_i^{(0)}}{2} + \partial_t^{(1)} f_i^{(0)} + \partial_t^{(0)} f_i^{(1)} \right) \\
 &+ Kn^3 \left(\frac{c_{i,\alpha}^3 \partial_\alpha^3 f_i^{(0)}}{6} + \frac{c_{i,\alpha}^2 \partial_\alpha^2 f_i^{(1)}}{2} + \frac{c_{i,\alpha}^2 \partial_\alpha^2 \partial_t^{(0)} f_i^{(0)}}{2} + c_{i,\alpha} \partial_\alpha f_i^{(2)} + \frac{c_{i,\alpha} \partial_\alpha (\partial_t^{(0)})^2 f_i^{(0)}}{2} + c_{i,\alpha} \partial_\alpha \partial_t^{(1)} f_i^{(0)} + c_{i,\alpha} \partial_\alpha \partial_t^{(0)} f_i^{(1)} + \frac{(\partial_t^{(0)})^3 f_i^{(0)}}{6} \right. \\
 &\left. + Kn^4 \left(\frac{c_{i,\alpha}^4 \partial_\alpha^4 f_i^{(0)}}{24} + \frac{c_{i,\alpha}^3 \partial_\alpha^3 f_i^{(1)}}{6} + \frac{c_{i,\alpha}^3 \partial_\alpha^3 \partial_t^{(0)} f_i^{(0)}}{6} + \frac{c_{i,\alpha}^2 \partial_\alpha^2 f_i^{(2)}}{2} + \frac{c_{i,\alpha}^2 \partial_\alpha^2 (\partial_t^{(0)})^2 f_i^{(0)}}{4} + \frac{c_{i,\alpha}^2 \partial_\alpha^2 \partial_t^{(1)} f_i^{(0)}}{2} + \frac{c_{i,\alpha}^2 \partial_\alpha^2 \partial_t^{(0)} f_i^{(1)}}{2} + c_{i,\alpha} \partial_\alpha f_i^{(3)} \right) \right)
 \end{aligned}$$

Splitting the equation in orders up to the order Kn^3 (References with High-Order expansion in Knudsen-order: Chai *et al.* [CHGS18]):

$$\begin{aligned}
 (Kn^{(0)}) : \quad & f_i^{(0)} = f_i^{eq}, \\
 (Kn^{(1)}) : \quad & (\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha) f_i^{(0)} = -\frac{f_i^{(1)}}{\tau}, \\
 (Kn^{(2)}) : \quad & \partial_t^{(1)} f_i^{(0)} + (\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha) f_i^{(1)} + \frac{(\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha)^2 f_i^{(0)}}{2} = -\frac{f_i^{(2)}}{\tau}, \\
 \text{ou} \\
 (Kn^{(2)}) : \quad & \partial_t^{(1)} f_i^{(0)} + (\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha) \left(1 - \frac{1}{2\tau}\right) f_i^{(1)} = -\frac{f_i^{(2)}}{\tau} \quad \rightarrow \quad f_i^{(1)} \text{ Formulation} \\
 \text{ou} \\
 (Kn^{(2)}) : \quad & \partial_t^{(1)} f_i^{(0)} + (\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha)^2 \left(\frac{1}{2} - \tau\right) f_i^{(0)} = -\frac{f_i^{(2)}}{\tau} \quad \rightarrow \quad f_i^{(0)} \text{ Formulation} \\
 (Kn^{(3)}) : \quad & \partial_t^{(2)} f_i^{(0)} + \partial_t^{(1)} f_i^{(1)} + \partial_t^{(1)} \left((\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha) f_i^{(0)} \right) + (\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha) f_i^{(2)} + \frac{(\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha)^2 f_i^{(1)}}{2} \\
 & + \frac{(\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha)^3 f_i^{(0)}}{6} = -\frac{f_i^{(3)}}{\tau}, \\
 \text{ou} \\
 (Kn^{(3)}) : \quad & \partial_t^{(2)} f_i^{(0)} + \left(2 - \frac{1}{\tau}\right) \partial_t^{(1)} f_i^{(1)} + \left(1 - \frac{1}{6\tau} - \tau\right) (\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha)^2 f_i^{(1)} = -\frac{f_i^{(3)}}{\tau}, \quad \rightarrow \quad f_i^{(1)} \text{ Formulation} \\
 \text{ou} \\
 (Kn^{(3)}) : \quad & \partial_t^{(2)} f_i^{(0)} + 2 \left(\frac{1}{2} - \tau\right) \partial_t^{(1)} (\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha) f_i^{(0)} + \left(\tau^2 - \tau + \frac{1}{6}\right) (\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha)^3 f_i^{(0)} = -\frac{f_i^{(3)}}{\tau}, \quad \rightarrow \quad f_i^{(0)} \text{ Formulation}
 \end{aligned} \tag{5.5}$$

Zero-Order Moment Balance

To retrieve the mass balance equation, we sum the Eq. (5.5) for $Kn^{(1)}$, $Kn^{(2)}$, and $Kn^{(3)}$ over $\sum_{i=0}$:

$$\begin{aligned}
 (Kn^{(1)}) : \quad & \sum_i \left(\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha \right) f_i^{(0)} = - \underbrace{\sum_i \left(\frac{f_i^{(1)}}{\tau} \right)}_{=0}, \\
 (Kn^{(1)}) : \quad & \partial_t^{(0)} \rho + \partial_\alpha (\rho u_\alpha) = 0, \\
 (Kn^{(2)}) : \quad & \sum_i \left[\partial_t^{(1)} f_i^{(0)} + \left(1 - \frac{1}{2\tau} \right) \left(\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha \right) f_i^{(1)} \right] = - \sum_i \left(\frac{f_i^{(2)}}{\tau} \right), \\
 (Kn^{(2)}) : \quad & \underbrace{\partial_t^{(1)} \sum_i f_i^{(0)}}_{=0} + \underbrace{\left(1 - \frac{1}{2\tau} \right) \partial_t^{(0)} \sum_i f_i^{(1)}}_{=0} + \underbrace{\left(1 - \frac{1}{2\tau} \right) \partial_\alpha \sum_i c_{i,\alpha} f_i^{(1)}}_{=0} = - \frac{1}{\tau} \sum_i f_i^{(2)} \\
 (Kn^{(2)}) : \quad & \partial_t^{(1)} \rho = 0,
 \end{aligned} \tag{5.6}$$

Extending the analysis up to $Kn^{(3)}$

$$\begin{aligned}
 (Kn^{(3)}) : \quad & \sum_i \left[\partial_t^{(2)} f_i^{(0)} + \left(2 - \frac{1}{\tau} \right) \partial_t^{(1)} f_i^{(1)} + \left(1 - \frac{1}{6\tau} - \tau \right) \left(\partial_t^{(0)} + c_{i,\alpha} \partial_\alpha \right)^2 f_i^{(1)} \right] = - \sum_i \left[\frac{f_i^{(3)}}{\tau} \right], \\
 (Kn^{(3)}) : \quad & \underbrace{\partial_t^{(2)} \sum_i f_i^{(0)}}_{=0} + \underbrace{\left(2 - \frac{1}{\tau} \right) \partial_t^{(1)} \sum_i f_i^{(1)}}_{=0} + \left(1 - \frac{1}{6\tau} - \tau \right) \left(\underbrace{\partial_t^{(0)} \partial_t^{(0)} \sum_i f_i^{(1)}}_{=0} + \underbrace{\partial_\alpha \partial_t^{(0)} \sum_i c_{i,\alpha} f_i^{(1)}}_{=0} + \underbrace{\partial_\beta \partial_t^{(0)} \sum_i c_{i,\beta} f_i^{(1)}}_{=0} \right) \\
 & + \left(1 - \frac{1}{6\tau} - \tau \right) \left(\partial_\alpha \partial_\beta \sum_i c_{i,\alpha} c_{i,\beta} f_i^{(1)} \right) = - \frac{1}{\tau} \underbrace{\sum_i f_i^{(3)}}_{=0}, \\
 (Kn^{(3)}) : \quad & \partial_t^{(2)} \rho + \left(1 - \frac{1}{6\tau} - \tau \right) \partial_\alpha \partial_\beta m_{\alpha\beta}^{(1)} = 0.
 \end{aligned} \tag{5.7}$$

First-Order Moment Balance

To retrieve the momentum balance equation, we sum the Eq. (5.5) for $Kn^{(1)}$, $Kn^{(2)}$, and $Kn^{(3)}$ over $\sum_{i=0} c_{i,\alpha}$:

$$\begin{aligned}
 (Kn^{(1)}) : \quad & \sum_i \left(\partial_t^{(0)} + c_{i,\beta} \partial_\beta \right) f_i^{(0)} c_{i,\alpha} = - \underbrace{\sum_i \left(\frac{f_i^{(1)}}{\tau} \right) c_{i,\alpha}}_{=0}, \\
 (Kn^{(1)}) : \quad & \partial_t^{(0)} (\rho u_\alpha) + \partial_\beta (\rho u_\alpha u_\beta + \rho c_s^2 \delta_{\alpha\beta}) = 0, \\
 (Kn^{(2)}) : \quad & \sum_i \left[\partial_t^{(1)} f_i^{(0)} + \left(1 - \frac{1}{2\tau} \right) \left(\partial_t^{(0)} + c_{i,\beta} \partial_\beta \right) f_i^{(1)} \right] c_{i,\alpha} = - \sum_i \left(\frac{f_i^{(2)}}{\tau} \right) c_{i,\alpha}, \\
 (Kn^{(2)}) : \quad & \underbrace{\partial_t^{(1)} \sum_i f_i^{(0)} c_{i,\alpha}}_{=0} + \underbrace{\left(1 - \frac{1}{2\tau} \right) \partial_t^{(0)} \sum_i f_i^{(1)} c_{i,\alpha}}_{=0} + \underbrace{\left(1 - \frac{1}{2\tau} \right) \partial_\beta \sum_i f_i^{(1)} c_{i,\alpha} c_{i,\beta}}_{=0} = - \frac{1}{\tau} \sum_i f_i^{(2)} c_{i,\alpha} \\
 (Kn^{(2)}) : \quad & \partial_t^{(1)} (\rho u_\alpha) + \left(1 - \frac{1}{2\tau} \right) \partial_\beta m_{\alpha\beta}^{(1)} = 0,
 \end{aligned} \tag{5.8}$$

Extending the analysis up to $Kn^{(3)}$

$$\begin{aligned}
 (Kn^{(3)}) : \quad & \sum_i \left[\partial_t^{(2)} f_i^{(0)} + \left(2 - \frac{1}{\tau}\right) \partial_t^{(1)} f_i^{(1)} + \left(1 - \frac{1}{6\tau} - \tau\right) \left(\partial_t^{(0)} + c_{i,\beta} \partial_\beta\right)^2 f_i^{(1)} \right] c_{i,\alpha} = - \sum_i \left[\frac{f_i^{(3)}}{\tau} \right] c_{i,\alpha}, \\
 (Kn^{(3)}) : \quad & \partial_t^{(2)} \sum_i f_i^{(0)} c_{i,\alpha} + \underbrace{\left(2 - \frac{1}{\tau}\right) \partial_t^{(1)} \sum_i f_i^{(1)} c_{i,\alpha}}_{=0} + \left(1 - \frac{1}{6\tau} - \tau\right) \underbrace{\left(\partial_t^{(0)} \partial_t^{(0)} \sum_i f_i^{(1)} c_{i,\alpha} \right)}_{=0} \\
 & + \left(1 - \frac{1}{6\tau} - \tau\right) \left(\partial_t^{(0)} \partial_\beta \sum_i f_i^{(1)} c_{i,\alpha} c_{i,\beta} + \partial_t^{(0)} \partial_\gamma \sum_i f_i^{(1)} c_{i,\alpha} c_{i,\gamma} + \partial_\beta \partial_\gamma \sum_i c_{i,\alpha} c_{i,\beta} c_{i,\gamma} f_i^{(1)} \right) = - \frac{1}{\tau} \underbrace{\sum_i f_i^{(3)}}_{=0}, \\
 (Kn^{(3)}) : \quad & \partial_t^{(2)} (\rho u_\alpha) + \left(1 - \frac{1}{6\tau} - \tau\right) \left(\partial_t^{(0)} \partial_\beta m_{\alpha\beta}^{(1)} + \partial_t^{(0)} \partial_\gamma m_{\alpha\gamma}^{(1)} + \partial_\beta \partial_\gamma m_{\alpha\beta\gamma}^{(1)} \right) = 0.
 \end{aligned} \tag{5.9}$$

Second-Order Non-Equilibrium Moment

To determine the unknown term $m_{\alpha\beta}^{(1)}$, we sum the Eq. (5.5) for $Kn^{(1)}$ over the second-order moment:

$$\begin{aligned}
 (Kn^{(1)}) : \quad & \sum_i \left(\partial_t^{(0)} + c_{i,\gamma} \partial_\gamma \right) f_i^{(0)} c_{i,\alpha} c_{i,\beta} = - \sum_i \left(\frac{f_i^{(1)}}{\tau} c_{i,\alpha} c_{i,\beta} \right), \\
 (Kn^{(1)}) : \quad & \partial_t^{(0)} (\rho u_\alpha u_\beta + \rho c_s^2 \delta_{\alpha\beta}) + \partial_\gamma (c_s^2 \rho (u_\alpha \delta_{\beta\gamma} + u_\beta \delta_{\alpha\gamma} + u_\gamma \delta_{\alpha\beta})) = - \frac{m_{\alpha\beta}^{(1)}}{\tau},
 \end{aligned}$$

notice that third-order moment for the lattice D2Q9 is not correct recovered. Expanding the terms and using Eqs. (5.6) and (5.8) for numerical manipulation:

$$\begin{aligned}
 (Kn^{(1)}) : \quad & \partial_t^{(0)} (\rho u_\alpha u_\beta + \rho c_s^2 \delta_{\alpha\beta}) + \partial_\gamma (c_s^2 \rho (u_\alpha \delta_{\beta\gamma} + u_\beta \delta_{\alpha\gamma} + u_\gamma \delta_{\alpha\beta})) = - \frac{m_{\alpha\beta}^{(1)}}{\tau}, \\
 (Kn^{(1)}) : \quad & \partial_t^{(0)} (\rho c_s^2 \delta_{\alpha\beta}) + \partial_t^{(0)} (\rho u_\alpha u_\beta) + c_s^2 (\partial_\gamma (\rho u_\alpha) \delta_{\beta\gamma} + \partial_\gamma (\rho u_\beta) \delta_{\alpha\gamma}) + \partial_\gamma (c_s^2 \rho u_\gamma \delta_{\alpha\beta}) = - \frac{m_{\alpha\beta}^{(1)}}{\tau}, \\
 (Kn^{(1)}) : \quad & c_s^2 \partial_t^{(0)} (\rho) \delta_{\alpha\beta} + \rho u_\alpha \partial_t^{(0)} (u_\beta) + u_\beta \partial_t^{(0)} (\rho u_\alpha) + c_s^2 (\partial_\beta (\rho u_\alpha) + \partial_\alpha (\rho u_\beta)) + c_s^2 \partial_\gamma (\rho u_\gamma) \delta_{\alpha\beta} = - \frac{m_{\alpha\beta}^{(1)}}{\tau}, \\
 (Kn^{(1)}) : \quad & -c_s^2 \partial_\gamma (\rho u_\gamma) \delta_{\alpha\beta} + \rho u_\alpha \partial_t^{(0)} (u_\beta) - u_\beta \partial_\gamma (\rho u_\alpha u_\gamma + \rho c_s^2 \delta_{\alpha\gamma}) + c_s^2 (\partial_\beta (\rho u_\alpha) + \partial_\alpha (\rho u_\beta)) + c_s^2 \partial_\gamma (\rho u_\gamma) \delta_{\alpha\beta} = - \frac{m_{\alpha\beta}^{(1)}}{\tau}, \\
 (Kn^{(1)}) : \quad & u_\alpha (-u_\beta \partial_t^{(0)} \rho - \partial_\gamma (\rho u_\beta u_\gamma + \rho c_s^2 \delta_{\beta\gamma})) - u_\beta \partial_\gamma (\rho u_\alpha u_\gamma + \rho c_s^2 \delta_{\alpha\gamma}) + c_s^2 (\partial_\beta (\rho u_\alpha) + \partial_\alpha (\rho u_\beta)) = - \frac{m_{\alpha\beta}^{(1)}}{\tau}, \\
 (Kn^{(1)}) : \quad & u_\alpha (u_\beta \partial_\gamma (\rho u_\gamma) - \partial_\gamma (\rho u_\beta u_\gamma + \rho c_s^2 \delta_{\beta\gamma})) - u_\beta \partial_\gamma (\rho u_\alpha u_\gamma + \rho c_s^2 \delta_{\alpha\gamma}) + c_s^2 (\partial_\beta (\rho u_\alpha) + \partial_\alpha (\rho u_\beta)) = - \frac{m_{\alpha\beta}^{(1)}}{\tau}, \\
 (Kn^{(1)}) : \quad & u_\alpha (-\rho u_\gamma \partial_\gamma u_\beta - c_s^2 \partial_\beta \rho) - u_\beta \partial_\gamma (\rho u_\alpha u_\gamma + \rho c_s^2 \delta_{\alpha\gamma}) + c_s^2 (\partial_\beta (\rho u_\alpha) + \partial_\alpha (\rho u_\beta)) = - \frac{m_{\alpha\beta}^{(1)}}{\tau}, \\
 (Kn^{(1)}) : \quad & -\rho u_\alpha u_\gamma \partial_\gamma u_\beta - c_s^2 u_\alpha \partial_\beta \rho - u_\beta \partial_\gamma \rho u_\alpha u_\gamma - u_\beta c_s^2 \partial_\alpha \rho + c_s^2 (\partial_\beta (\rho u_\alpha) + \partial_\alpha (\rho u_\beta)) = - \frac{m_{\alpha\beta}^{(1)}}{\tau}, \\
 (Kn^{(1)}) : \quad & -\partial_\gamma (\rho u_\alpha u_\beta u_\gamma) - c_s^2 (u_\alpha \partial_\beta \rho + u_\beta \partial_\alpha \rho) + c_s^2 (\partial_\beta (\rho u_\alpha) + \partial_\alpha (\rho u_\beta)) = - \frac{m_{\alpha\beta}^{(1)}}{\tau}, \\
 (Kn^{(1)}) : \quad & -\partial_\gamma (\rho u_\alpha u_\beta u_\gamma) + c_s^2 \rho (\partial_\beta u_\alpha + \partial_\alpha u_\beta) = - \frac{m_{\alpha\beta}^{(1)}}{\tau}, \quad \rightarrow \quad m_{\alpha\beta}^{(1)} = -c_s^2 \tau \rho (\partial_\beta u_\alpha + \partial_\alpha u_\beta) + \tau \partial_\gamma (\rho u_\alpha u_\beta u_\gamma)
 \end{aligned} \tag{5.10}$$

Balance Equations

To recover the mass balance equation, we sum the Eqs. (5.6), (5.7), and (5.7):

$$\begin{aligned} \partial_t \rho + \partial_\alpha (\rho u_\alpha) + \underbrace{\left(1 - \frac{1}{2\tau}\right) \partial_\beta m_{\alpha\beta}^{(1)} + \dots + \mathcal{O}(Kn^{(\geq 3)})}_{\text{Numerical Errors}} &= 0, \\ \partial_t \rho + \partial_\alpha (\rho u_\alpha) + \underbrace{\left(\tau - \frac{1}{2}\right) (\partial_\beta (-c_s^2 \rho (\partial_\beta u_\alpha + \partial_\alpha u_\beta) + \partial_\gamma (\rho u_\alpha u_\beta u_\gamma))) + \dots + \mathcal{O}(Kn^{(>3)})}_{\text{Numerical Errors}} &= 0, \end{aligned}$$

Repeating the process for momentum balance, we sum the Eqs. (5.8), (5.9), (5.9), and (5.10):

$$\begin{aligned} \partial_t \rho u_\alpha + \partial_\beta (\rho u_\alpha \rho u_\beta) &= \partial_\beta \left(-\rho c_s^2 \delta_{\alpha\beta} + \left(\tau - \frac{1}{2}\right) \rho c_s^2 (\partial_\alpha u_\beta + \partial_\beta u_\alpha) \right) \\ &\quad - \underbrace{\left(\tau - \frac{1}{2}\right) \partial_\gamma (\rho u_\alpha u_\beta u_\gamma) - \left(1 - \frac{1}{6\tau} - \tau\right) (\partial_t^{(0)} \partial_\beta m_{\alpha\beta}^{(1)} + \partial_t^{(0)} \partial_\gamma m_{\alpha\gamma}^{(1)} + \partial_\beta \partial_\gamma m_{\alpha\beta\gamma}^{(1)}) + \dots + \mathcal{O}(Kn^{(>3)})}_{\text{Numerical Errors}} = 0, \\ \partial_t \rho u_\alpha + \partial_\beta (\rho u_\alpha \rho u_\beta) &= \partial_\beta \left(-p \delta_{\alpha\beta} + \mu (\partial_\alpha u_\beta + \partial_\beta u_\alpha) \right) \\ &\quad - \underbrace{\left(\tau - \frac{1}{2}\right) \partial_\gamma (\rho u_\alpha u_\beta u_\gamma) - \left(1 - \frac{1}{6\tau} - \tau\right) (\partial_t^{(0)} \partial_\beta m_{\alpha\beta}^{(1)} + \partial_t^{(0)} \partial_\gamma m_{\alpha\gamma}^{(1)} + \partial_\beta \partial_\gamma m_{\alpha\beta\gamma}^{(1)}) + \dots + \mathcal{O}(Kn^{(>3)})}_{\text{Numerical Errors}} = 0, \end{aligned}$$

where $\mu = \rho c_s^2 (\tau - 1/2)$.

5.1.3 Lattice Direction Moments: (Multi-Speed Lattices) D2V17 Velocity-Space Discretization



Raw Moments

$$\begin{aligned} \underbrace{\sum_{i=0} f_i^{eq}}_{\text{Zero-Order Moment}} &= \sum_{i=0} f_i = \rho, & \underbrace{\sum_{i=0} f_i^{eq} c_{i,x}}_{\text{x-First-Order Moment}} &= \rho u_x^*, & \underbrace{\sum_{i=0} f_i^{eq} c_{i,y}}_{\text{y-First-Order Moment}} &= \rho u_y^* \\ \underbrace{\sum_{i=0} f_i^{eq} c_{i,x} c_{i,x}}_{\text{xx-Second-Order Moment}} &= \rho c_s^2 + \rho (u_x^*)^2, & \underbrace{\sum_{i=0} f_i^{eq} c_{i,x} c_{i,y}}_{\text{xy-Second-Order Moment}} &= \rho u_x^* u_y^*, & \underbrace{\sum_{i=0} f_i^{eq} c_{i,y} c_{i,y}}_{\text{yy-Second-Order Moment}} &= \rho c_s^2 + \rho (u_y^*)^2 \end{aligned}$$

$$\underbrace{\sum_{i=0} f_i^{eq} c_{i,x} c_{i,x} c_{i,x}}_{\text{xxx-Third-Order Moment}} = 3\rho c_s^2 u_x^* + \rho (u_x^*)^3, \quad \underbrace{\sum_{i=0} f_i^{eq} c_{i,x} c_{i,x} c_{i,y}}_{\text{xyy-Third-Order Moment}} = \rho c_s^2 u_y^* + \rho (u_x^*)^2 u_y^*,$$

$$\underbrace{\sum_{i=0} f_i^{eq} c_{i,x} c_{i,y} c_{i,y}}_{\text{xyy-Third-Order Moment}} = \rho c_s^2 u_x^* + \rho u_x^* (u_y^*)^2, \quad \underbrace{\sum_{i=0} f_i^{eq} c_{i,y} c_{i,y} c_{i,y}}_{\text{yyy-Third-Order Moment}} = 3\rho c_s^2 u_y^* + \rho (u_y^*)^3$$

Hermite Moments

$$\begin{aligned} \underbrace{\sum_{i=0} f_i^{eq}}_{\text{Zero-Order Hermite Moment}} &= \sum_{i=0} f_i = \rho, & \underbrace{\sum_{i=0} f_i^{eq} c_{i,x}}_{\text{x-First-Order Hermite Moment}} &= \rho u_x^*, & \underbrace{\sum_{i=0} f_i^{eq} c_{i,y}}_{\text{y-First-Order Hermite Moment}} &= \rho u_y^* \\ \underbrace{\sum_{i=0} f_i^{eq} c_{i,x} c_{i,x}}_{\text{xx-Second-Order Moment}} &= \rho (u_x^*)^2, & \underbrace{\sum_{i=0} f_i^{eq} c_{i,x} c_{i,y}}_{\text{xy-Second-Order Moment}} &= \rho u_x^* u_y^*, & \underbrace{\sum_{i=0} f_i^{eq} c_{i,y} c_{i,y}}_{\text{yy-Second-Order Moment}} &= \rho (u_y^*)^2 \\ \underbrace{\sum_{i=0} f_i^{eq} c_{i,x} c_{i,x} c_{i,x}}_{\text{xxx-Third-Order Moment}} &= \rho (u_x^*)^3, & \underbrace{\sum_{i=0} f_i^{eq} c_{i,x} c_{i,x} c_{i,y}}_{\text{xyy-Third-Order Moment}} &= \rho (u_x^*)^2 u_y^*, \\ \underbrace{\sum_{i=0} f_i^{eq} c_{i,x} c_{i,y} c_{i,y}}_{\text{xyy-Third-Order Moment}} &= \rho u_x^* (u_y^*)^2, & \underbrace{\sum_{i=0} f_i^{eq} c_{i,y} c_{i,y} c_{i,y}}_{\text{yyy-Third-Order Moment}} &= \rho (u_y^*)^3 \end{aligned}$$

Doing the Chapman-Enskog analysis for the lattice D2V17, we obtain the same Eqs. (5.6), (5.7), (5.7), (5.8), (5.9), and (5.9). However, the calculation of the second-order moment of non-equilibrium $m_{\alpha\beta}^{(1)}$ will change due to the D2V17 be able to correctly recover the third-order moments:

$$\begin{aligned} (Kn^{(1)}): \quad & \sum_i \left(\partial_t^{(0)} + c_{i,\gamma} \partial_\gamma \right) f_i^{(0)} c_{i,\alpha} c_{i,\beta} = - \sum_i \left(\frac{f_i^{(1)}}{\tau} c_{i,\alpha} c_{i,\beta} \right), \\ (Kn^{(1)}): \quad & \partial_t^{(0)} (\rho u_\alpha u_\beta + \rho c_s^2 \delta_{\alpha\beta}) + \partial_\gamma (c_s^2 \rho (u_\alpha \delta_{\beta\gamma} + u_\beta \delta_{\alpha\gamma} + u_\gamma \delta_{\alpha\beta})) = - \frac{m_{\alpha\beta}^{(1)}}{\tau}, \\ (Kn^{(1)}): \quad & \partial_t^{(0)} (\rho u_\alpha u_\beta + \rho c_s^2 \delta_{\alpha\beta}) + \partial_\gamma (c_s^2 \rho (u_\alpha \delta_{\beta\gamma} + u_\beta \delta_{\alpha\gamma} + u_\gamma \delta_{\alpha\beta})) + \partial_\gamma (u_\alpha u_\beta u_\gamma) = - \frac{m_{\alpha\beta}^{(1)}}{\tau}, \\ (Kn^{(1)}): \quad & \partial_t^{(0)} (\rho c_s^2 \delta_{\alpha\beta}) + \partial_t^{(0)} (\rho u_\alpha u_\beta) + c_s^2 (\partial_\gamma (\rho u_\alpha) \delta_{\beta\gamma} + \partial_\gamma (\rho u_\beta) \delta_{\alpha\gamma}) + \partial_\gamma (c_s^2 \rho u_\gamma \delta_{\alpha\beta}) + \partial_\gamma (u_\alpha u_\beta u_\gamma) = - \frac{m_{\alpha\beta}^{(1)}}{\tau}, \\ (Kn^{(1)}): \quad & c_s^2 \partial_t^{(0)} (\rho) \delta_{\alpha\beta} + \rho u_\alpha \partial_t^{(0)} (u_\beta) + u_\beta \partial_t^{(0)} (\rho u_\alpha) + c_s^2 (\partial_\beta (\rho u_\alpha) + \partial_\alpha (\rho u_\beta)) + c_s^2 \partial_\gamma (\rho u_\gamma) \delta_{\alpha\beta} + \partial_\gamma (u_\alpha u_\beta u_\gamma) = - \frac{m_{\alpha\beta}^{(1)}}{\tau}, \\ (Kn^{(1)}): \quad & -c_s^2 \partial_\gamma (\rho u_\gamma) \delta_{\alpha\beta} + \rho u_\alpha \partial_t^{(0)} (u_\beta) - u_\beta \partial_\gamma (\rho u_\alpha u_\gamma + \rho c_s^2 \delta_{\alpha\gamma}) + c_s^2 (\partial_\beta (\rho u_\alpha) + \partial_\alpha (\rho u_\beta)) + c_s^2 \partial_\gamma (\rho u_\gamma) \delta_{\alpha\beta} + \partial_\gamma (u_\alpha u_\beta u_\gamma) = - \frac{m_{\alpha\beta}^{(1)}}{\tau}, \\ (Kn^{(1)}): \quad & u_\alpha (-u_\beta \partial_t^{(0)} \rho - \partial_\gamma (\rho u_\beta u_\gamma + \rho c_s^2 \delta_{\beta\gamma})) - u_\beta \partial_\gamma (\rho u_\alpha u_\gamma + \rho c_s^2 \delta_{\alpha\gamma}) + c_s^2 (\partial_\beta (\rho u_\alpha) + \partial_\alpha (\rho u_\beta)) + \partial_\gamma (u_\alpha u_\beta u_\gamma) = - \frac{m_{\alpha\beta}^{(1)}}{\tau}, \\ (Kn^{(1)}): \quad & u_\alpha (u_\beta \partial_\gamma (\rho u_\gamma) - \partial_\gamma (\rho u_\beta u_\gamma + \rho c_s^2 \delta_{\beta\gamma})) - u_\beta \partial_\gamma (\rho u_\alpha u_\gamma + \rho c_s^2 \delta_{\alpha\gamma}) + c_s^2 (\partial_\beta (\rho u_\alpha) + \partial_\alpha (\rho u_\beta)) + \partial_\gamma (u_\alpha u_\beta u_\gamma) = - \frac{m_{\alpha\beta}^{(1)}}{\tau}, \\ (Kn^{(1)}): \quad & u_\alpha (-\rho u_\gamma \partial_\gamma u_\beta - c_s^2 \partial_\beta \rho) - u_\beta \partial_\gamma (\rho u_\alpha u_\gamma + \rho c_s^2 \delta_{\alpha\gamma}) + c_s^2 (\partial_\beta (\rho u_\alpha) + \partial_\alpha (\rho u_\beta)) + \partial_\gamma (u_\alpha u_\beta u_\gamma) = - \frac{m_{\alpha\beta}^{(1)}}{\tau}, \\ (Kn^{(1)}): \quad & -\rho u_\alpha u_\gamma \partial_\gamma u_\beta - c_s^2 u_\alpha \partial_\beta \rho - u_\beta \partial_\gamma \rho u_\alpha u_\gamma - u_\beta c_s^2 \partial_\alpha \rho + c_s^2 (\partial_\beta (\rho u_\alpha) + \partial_\alpha (\rho u_\beta)) + \partial_\gamma (u_\alpha u_\beta u_\gamma) = - \frac{m_{\alpha\beta}^{(1)}}{\tau}, \\ (Kn^{(1)}): \quad & -\partial_\gamma (\rho u_\alpha u_\beta u_\gamma) - c_s^2 (u_\alpha \partial_\beta \rho + u_\beta \partial_\alpha \rho) + c_s^2 (\partial_\beta (\rho u_\alpha) + \partial_\alpha (\rho u_\beta)) + \partial_\gamma (u_\alpha u_\beta u_\gamma) = - \frac{m_{\alpha\beta}^{(1)}}{\tau}, \\ (Kn^{(1)}): \quad & c_s^2 \rho (\partial_\beta u_\alpha + \partial_\alpha u_\beta) = - \frac{m_{\alpha\beta}^{(1)}}{\tau}, \quad \rightarrow \quad m_{\alpha\beta}^{(1)} = -c_s^2 \tau \rho (\partial_\beta u_\alpha + \partial_\alpha u_\beta) \end{aligned} \tag{5.11}$$

Repeating the process for momentum balance, we sum the Eqs. (5.8), (5.9), (5.9), and (5.11):

$$\begin{aligned}
 \partial_t \rho u_\alpha + \partial_\beta (\rho u_\alpha \rho u_\beta) &= \partial_\beta \left(-\rho c_s^2 \delta_{\alpha\beta} + \left(\tau - \frac{1}{2} \right) \rho c_s^2 (\partial_\alpha u_\beta + \partial_\beta u_\alpha) \right) \\
 &\quad - \underbrace{\left(1 - \frac{1}{6\tau} - \tau \right) \left(\partial_t^{(0)} \partial_\beta m_{\alpha\beta}^{(1)} + \partial_t^{(0)} \partial_\gamma m_{\alpha\gamma}^{(1)} + \partial_\beta \partial_\gamma m_{\alpha\beta\gamma}^{(1)} \right) + \dots + \mathcal{O}(Kn^{(>3)})}_{\text{Numerical Errors}} = 0, \\
 \partial_t \rho u_\alpha + \partial_\beta (\rho u_\alpha \rho u_\beta) &= \partial_\beta \left(-p \delta_{\alpha\beta} + \mu (\partial_\alpha u_\beta + \partial_\beta u_\alpha) \right) \\
 &\quad - \underbrace{\left(1 - \frac{1}{6\tau} - \tau \right) \left(\partial_t^{(0)} \partial_\beta m_{\alpha\beta}^{(1)} + \partial_t^{(0)} \partial_\gamma m_{\alpha\gamma}^{(1)} + \partial_\beta \partial_\gamma m_{\alpha\beta\gamma}^{(1)} \right) + \dots + \mathcal{O}(Kn^{(>3)})}_{\text{Numerical Errors}} = 0,
 \end{aligned}$$

where $\mu = \rho c_s^2 (\tau - 1/2)$.

5.2 Tutorials

DIFFERENCE FINITE SCHEME DERIVED FROM MRT-HERMITE LATTICE BOLTZMANN FOR CONVECTIVE-DIFFUSIVE EQUATION IN D1Q5ZOT

6.1 MRT Lattice Boltzmann equation for Convective-Diffusive Equation

To describe convective-diffusive problems, the MRT lattice Boltzmann formulation is described by:

$$f_i(x_\alpha + e_{i,\alpha}\delta t, t + \delta t) = f_i(x_\alpha, t) - (\mathbf{M}^{-1}\mathbf{S}\mathbf{M})_{ik} (f_k - f_k^{eq}), \quad (6.1)$$

where $\mathbf{M}$ is matrix of moments, $\mathbf{S}$ is the diagonal matrix of relaxation times, and $\mathbf{M}^{-1}$ is the inverse matrix of $\mathbf{M}$. Considering the problem description in a lattice D1Q5ZOT, we have:

$$\mathbf{M} = \begin{pmatrix} 1 & e_{i,x} & e_{i,x}^2 - c_s^2 & e_{i,x}^3 - 3e_{i,x}c_s^2 & e_{i,x}^4 - 6e_{i,x}^2c_s^2 + 3c_s^4 \\ 0 & 1 & -c_s^2 & 0 & 3c_s^4 \\ 0 & 1 - c_s^2 & 1 - c_s^2 & 3c_s^2 - 1 & 3c_s^4 - 6c_s^2 + 1 \\ 0 & 1 - 3c_s^2 & 3c_s^2 - 1 & 27 - 9c_s^2 & 3c_s^4 - 6c_s^2 + 1 \\ 3c_s^4 & 3c_s^4 - 6c_s^2 + 1 & 3c_s^4 - 6c_s^2 + 1 & 3c_s^4 - 54c_s^2 + 81 & 3c_s^4 - 54c_s^2 + 81 \end{pmatrix}, \quad \mathbf{S} = \begin{pmatrix} s_0 & 0 & 0 & 0 & 0 \\ 0 & s_1 & 0 & 0 & 0 \\ 0 & 0 & s_2 & 0 & 0 \\ 0 & 0 & 0 & s_3 & 0 \\ 0 & 0 & 0 & 0 & s_4 \end{pmatrix}$$

The equilibrium distribution function is defined by

$$\begin{aligned} f_i^{eq} &= w_i \sum_{\theta=0}^4 \frac{a^\theta \mathcal{H}^\theta}{\theta! c_s^{2\theta}} = w_i \left(u \mathcal{H}^0 + \frac{B_\alpha \mathcal{H}^1}{c_s^2} + \frac{(\Lambda_{\alpha\beta} + \Lambda_{\alpha\beta}^{aux} - c_s^2 u \delta_{\alpha\beta}) \mathcal{H}^2}{2c_s^4} + \frac{(\Pi_{\alpha\beta\gamma} + \Pi_{\alpha\beta\gamma}^{aux} - \tilde{3}c_s^2 B_\alpha \delta_{\beta\gamma}) \mathcal{H}^3}{6c_s^6} + \frac{(\Upsilon_{\alpha\beta\gamma\delta} + \Upsilon_{\alpha\beta\gamma\delta}^{aux}) \mathcal{H}^4}{24c_s^8} \right) \\ &= w_i \left(u + \frac{e_{i,\alpha} B_\alpha}{c_s^2} + \frac{(\Lambda_{\alpha\beta} + \Lambda_{\alpha\beta}^{aux} - c_s^2 u \delta_{\alpha\beta}) (e_{i,\alpha} e_{i,\beta} - c_s^2 \delta_{\alpha\beta})}{2c_s^4} + \frac{(\Pi_{\alpha\beta\gamma} + \Pi_{\alpha\beta\gamma}^{aux} - \tilde{3}c_s^2 B_\alpha \delta_{\beta\gamma}) (e_{i,\alpha} e_{i,\beta} e_{i,\gamma} - \tilde{3}c_s^2 e_{i,\alpha} \delta_{\beta\gamma})}{6c_s^6} \right) \end{aligned}$$

where $\Lambda_{\alpha\beta} = c_s^2 C \delta_{\alpha\beta}$, C is the Diffusive term, $\Upsilon = (16 + 5\sqrt{10})/6$ is a correction term to correct recover the fourth-order moment, and $\Lambda_{\alpha\beta}^{aux}$ is a correction term defined by

$$\Lambda_{\alpha\beta}^{aux} = \int B'_\alpha B'_\beta du = \int \left(\frac{\partial B_\alpha}{\partial u} \frac{\partial B_\beta}{\partial u} \right) du.$$

```
# Only to check if the moments are correctly recovered
# display(simplify(cx*0+1))
# display(simplify(cx))
# display(simplify(cx*cx - cs2))
# display(simplify(cx*cx*cx - 3*cx*cs2))
```

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```
# display(simplify(cx*cx*cx*cx - 6*cx*cs2 + 3*cs2**2))
# display(simplify(sum(feq)))
# display(simplify(sum(feq*cx)))
# display(simplify(sum(feq*cx*cx)))
# display(simplify(sum(feq*cx*cx*cx)))
# display(simplify(sum(feq*cx*cx*cx*cx)))
```

6.1.1 Raw Moments

```
a0=simplify(sum(feq))
ax=simplify(sum(feq*cx))
axx=simplify(sum(feq*cx*cx))
axxx=simplify(sum(feq*cx*cx*cx))
axxxx=simplify(sum(feq*cx*cx*cx*cx))
display(Math(r"\begin{gathered}"
  r"\underbrace{\sum_{i=0} f_{i}^{eq} = \sum_{i=0} f_{i}}_{\text{Zero-}
  \rightarrow \text{Order Moment}} = " + sp.latex(a0)
  +r", \quad \quad \underbrace{\sum_{i=0} f_{i}^{eq} e_{i,x}}_{\text{x-}
  \rightarrow \text{First-Order Moment}} = " + sp.latex(ax)
  +r", \quad \quad \underbrace{\sum_{i=0} f_{i}^{eq} e_{i,x} e_{i,x}}_{\text{xx-}
  \rightarrow \text{Second-Order Moment}} = " + sp.latex(axx)
  +r".\end{gathered}"
))
display(Math(r"\begin{gathered}"
  r"\underbrace{\sum_{i=0} f_{i}^{eq} e_{i,x} e_{i,x} e_{i,x}}_{\text{xxx-}
  \rightarrow \text{Third-Order Moment}} = " + sp.latex(axxx)
  +r", \quad \quad \underbrace{\sum_{i=0} f_{i}^{eq} e_{i,x} e_{i,x} e_{i,x} e_{i,x}}_{\text{xxxx-}
  \rightarrow \text{Fourth-Order Moment}} = " + sp.latex(axxxx)
  +r".\end{gathered}"
))
```

$$\begin{aligned}
 \underbrace{\sum_{i=0} f_i^{eq} = \sum_{i=0} f_i}_{\text{Zero-Order Moment}} &= u, & \underbrace{\sum_{i=0} f_i^{eq} e_{i,x}}_{\text{x-First-Order Moment}} &= B_\alpha, & \underbrace{\sum_{i=0} f_i^{eq} e_{i,x} e_{i,x}}_{\text{xx-Second-Order Moment}} &= \Lambda_{\alpha\beta}. \\
 \underbrace{\sum_{i=0} f_i^{eq} e_{i,x} e_{i,x} e_{i,x}}_{\text{xxx-Third-Order Moment}} &= \Pi_{\alpha\beta\gamma}, & \underbrace{\sum_{i=0} f_i^{eq} e_{i,x} e_{i,x} e_{i,x} e_{i,x}}_{\text{xxxx-Fourth-Order Moment}} &= \Gamma_{\alpha\beta\gamma\delta}.
 \end{aligned}$$

6.1.2 Hermite Moments

```
aH0=simplify(sum(feq))
aHx=simplify(sum(feq*cx))
aHxx=simplify(sum(feq*(cx*cx - cs**2)))
aHxxx=simplify(sum(feq*(cx*cx*cx - 3*cx*cs**2)))
aHxxxx=simplify(sum(feq*(cx*cx*cx*cx - 6*cx*cx*cs**2 + 3*cs**4)))
display(Math(r"\begin{gathered}"
  r"\underbrace{\sum_{i=0} f_{i}^{eq} = \sum_{i=0} f_{i}}_{\text{Zero-}
  \rightarrow \text{Order Hermite Moment}} = " + sp.latex(aH0)
  +r", \quad \quad \underbrace{\sum_{i=0} f_{i}^{eq} e_{i,x}}_{\text{x-}
  \rightarrow \text{First-Order Hermite Moment}} = " + sp.latex(aHx)
  +r".\end{gathered}"
))
```

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```

+r",\quad \quad \underbrace{\sum_{i=0} f_{\{i\}}^{eq} \left( e_{\{i,x\}} e_{\{i,x\}} - \right.}
\leftarrow c_{\{s\}}^{\{2\}} \left. \right) }_{\text{\texttt{xx-Second-Order Hermite Moment}}} =" + sp.latex(aHxx)
+r".\end{gathered}"
))
display(Math(r"\begin{gathered}"
+r"\underbrace{\sum_{i=0} f_{\{i\}}^{eq} \left( e_{\{i,x\}} e_{\{i,x\}} e_{\{i,x\}} - 3c_{\{s\}}^{\{2\}} e_{\{i,x\}} \right) }_{\text{\texttt{xxx-Third-Order Hermite Moment}}} =" + sp.
\leftarrow latex(aHxxx)
+r",\quad \quad \underbrace{\sum_{i=0} f_{\{i\}}^{eq} \left( e_{\{i,x\}} e_{\{i,x\}} e_{\{i,x\}} e_{\{i,x\}} - 6c_{\{s\}}^{\{2\}} e_{\{i,x\}} e_{\{i,x\}} + 3c_{\{s\}}^{\{4\}} \right) }_{\text{\texttt{xxxx-Fourth-Order Hermite Moment}}} =" + sp.latex(aHxxxx)
+r".\end{gathered}"
))

```

$$\begin{aligned}
\underbrace{\sum_{i=0} f_i^{eq} = \sum_{i=0} f_i}_{\text{Zero-Order Hermite Moment}} &= u, & \underbrace{\sum_{i=0} f_i^{eq} e_{i,x}}_{\text{x-First-Order Hermite Moment}} &= B_\alpha, & \underbrace{\sum_{i=0} f_i^{eq} (e_{i,x} e_{i,x} - c_s^2)}_{\text{xx-Second-Order Hermite Moment}} &= \Lambda_{\alpha\beta} - c_s^2 u. \\
\underbrace{\sum_{i=0} f_i^{eq} (e_{i,x} e_{i,x} e_{i,x} - 3c_s^2 e_{i,x})}_{\text{xxx-Third-Order Hermite Moment}} &= -3B_\alpha c_s^2 + \Pi_{\alpha\beta\gamma}, & \underbrace{\sum_{i=0} f_i^{eq} (e_{i,x} e_{i,x} e_{i,x} e_{i,x} - 6c_s^2 e_{i,x} e_{i,x} + 3c_s^4)}_{\text{xxxx-Fourth-Order Hermite Moment}} &= \Gamma_{\alpha\beta\gamma\delta} - 6\Lambda_{\alpha\beta} c_s^2 + 3c_s^4 u.
\end{aligned}$$

6.2 FDM Interpretation in MRT Framework

Reinterpreting the Eq. (??) with a shift in distribution function direction:

$$f_{i,x}^{t+1} = f_{i,x-e_i}^t - (\mathbf{M}^{-1} \mathbf{S} \mathbf{M})_{ik} \left(f_{k,x-e_i}^t - f_{k,x-e_i}^{eq,t} \right) = (\mathbf{M}^{-1} (\mathbf{I} - \mathbf{S}) \mathbf{M})_{ik} f_{k,x-e_i}^t + (\mathbf{M}^{-1} \mathbf{S} \mathbf{M})_{ik} f_{k,x-e_i}^{eq,t},$$

where $f_i(x_\alpha + e_{i,\alpha} \delta t, t + \delta t) = f_{i,x+c_i}^{t+1}$ for simplicate the notation and $\mathbf{I}$ is identity matrix. Defining the shift operator $T_{\Delta x}^{c_i}$, where $f_{k,x-e_i}^t = T_{\Delta x}^{c_i} [f_{k,x}^t]$, we can rewrite the above equation inside of the moments space:

$$\mathbf{m}_{k,x}^{t+1} = \mathbf{P} \mathbf{m}_{k,x}^t + \mathbf{Q} \mathbf{m}_{k,x}^{eq,t}, \quad (6.2)$$

where $\mathbf{m}_{k,x}^t = M_{ik} f_{i,x}^t$, k is the index of moments in the space moment, $\mathbf{T} := \mathbf{M}(\text{diag}(T_{\Delta x}^{c_0}, T_{\Delta x}^{c_1}, T_{\Delta x}^{c_2}, T_{\Delta x}^{c_3}, T_{\Delta x}^{c_4})) \mathbf{M}^{-1}$, $\mathbf{P} := \mathbf{T}(\mathbf{I} - \mathbf{S})$, and $\mathbf{Q} := \mathbf{T} \mathbf{S}$. The matrices are given by:

$$\mathbf{T} = \begin{bmatrix} \frac{T_{\Delta x}^{c_0} (3c_s^4 - 10c_s^2 + 9)}{9} - \frac{3T_{\Delta x}^{c_1} c_s^2 (c_s^2 - 3)}{16} - \frac{3T_{\Delta x}^{c_2} c_s^2 (c_s^2 - 3)}{16} + \frac{T_{\Delta x}^{c_3} c_s^2 (3c_s^2 - 1)}{144} + \frac{T_{\Delta x}^{c_4} c_s^2 (3c_s^2 - 1)}{144} \\ \frac{c_s^2 (-9T_{\Delta x}^{c_1} (c_s^2 - 3) + 9T_{\Delta x}^{c_2} (c_s^2 - 3) + T_{\Delta x}^{c_3} (3c_s^2 - 1) - T_{\Delta x}^{c_4} (3c_s^2 - 1))}{48} \\ \frac{c_s^2 (-16T_{\Delta x}^{c_0} (3c_s^4 - 10c_s^2 + 9) + 27T_{\Delta x}^{c_1} (c_s^2 - 3)(c_s^2 - 1) + 27T_{\Delta x}^{c_2} (c_s^2 - 3)(c_s^2 - 1) - T_{\Delta x}^{c_3} (c_s^2 - 9)(3c_s^2 - 1) - T_{\Delta x}^{c_4} (c_s^2 - 9)(3c_s^2 - 1))}{144} \\ \frac{c_s^2 (c_s^2 - 3)(3c_s^2 - 1)(3T_{\Delta x}^{c_1} - 3T_{\Delta x}^{c_2} - T_{\Delta x}^{c_3} + T_{\Delta x}^{c_4})}{16} \\ \frac{c_s^2 (16T_{\Delta x}^{c_0} c_s^2 (3c_s^4 - 10c_s^2 + 9) - 9T_{\Delta x}^{c_1} (c_s^2 - 3)(3c_s^4 - 6c_s^2 + 1) - 9T_{\Delta x}^{c_2} (c_s^2 - 3)(3c_s^4 - 6c_s^2 + 1) + T_{\Delta x}^{c_3} (3c_s^2 - 1)(c_s^4 - 18c_s^2 + 27) + T_{\Delta x}^{c_4} (3c_s^2 - 1)(c_s^4 - 18c_s^2 + 27))}{48} \end{bmatrix}$$

$$\mathbf{P} = \begin{bmatrix} -\frac{(s_0 - 1)(16T_{\Delta x}^{c_0} (3c_s^4 - 10c_s^2 + 9) - 27T_{\Delta x}^{c_1} c_s^2 (c_s^2 - 3) - 27T_{\Delta x}^{c_2} c_s^2 (c_s^2 - 3) + T_{\Delta x}^{c_3} c_s^2 (3c_s^2 - 1) + T_{\Delta x}^{c_4} c_s^2 (3c_s^2 - 1))}{144} \\ \frac{c_s^2 (s_0 - 1)(9T_{\Delta x}^{c_1} (c_s^2 - 3) - 9T_{\Delta x}^{c_2} (c_s^2 - 3) - T_{\Delta x}^{c_3} (3c_s^2 - 1) + T_{\Delta x}^{c_4} (3c_s^2 - 1))}{48} \\ \frac{c_s^2 (s_0 - 1)(16T_{\Delta x}^{c_0} (3c_s^4 - 10c_s^2 + 9) - 27T_{\Delta x}^{c_1} (c_s^2 - 3)(c_s^2 - 1) - 27T_{\Delta x}^{c_2} (c_s^2 - 3)(c_s^2 - 1) + T_{\Delta x}^{c_3} (c_s^2 - 9)(3c_s^2 - 1) + T_{\Delta x}^{c_4} (c_s^2 - 9)(3c_s^2 - 1))}{144} \\ -\frac{c_s^2 (c_s^2 - 3)(3c_s^2 - 1)(s_0 - 1)(3T_{\Delta x}^{c_1} - 3T_{\Delta x}^{c_2} - T_{\Delta x}^{c_3} + T_{\Delta x}^{c_4})}{16} \\ -\frac{c_s^2 (s_0 - 1)(16T_{\Delta x}^{c_0} c_s^2 (3c_s^4 - 10c_s^2 + 9) - 9T_{\Delta x}^{c_1} (c_s^2 - 3)(3c_s^4 - 6c_s^2 + 1) - 9T_{\Delta x}^{c_2} (c_s^2 - 3)(3c_s^4 - 6c_s^2 + 1) + T_{\Delta x}^{c_3} (3c_s^2 - 1)(c_s^4 - 18c_s^2 + 27) + T_{\Delta x}^{c_4} (3c_s^2 - 1)(c_s^4 - 18c_s^2 + 27))}{48} \end{bmatrix}$$

$$\mathbf{Q} = \begin{bmatrix} \frac{s_0(16T_{\Delta x}^{c_0}(3c_s^4-10c_s^2+9)-27T_{\Delta x}^{c_1}c_s^2(c_s^2-3)-27T_{\Delta x}^{c_2}c_s^2(c_s^2-3)+T_{\Delta x}^{c_3}c_s^2(3c_s^2-1)+T_{\Delta x}^{c_4}c_s^2(3c_s^2-1))}{144} \\ -\frac{c_s^2s_0(9T_{\Delta x}^{c_1}(c_s^2-3)-9T_{\Delta x}^{c_2}(c_s^2-3)-T_{\Delta x}^{c_3}(3c_s^2-1)+T_{\Delta x}^{c_4}(3c_s^2-1))}{48} \\ -\frac{c_s^2s_0(16T_{\Delta x}^{c_0}(3c_s^4-10c_s^2+9)-27T_{\Delta x}^{c_1}(c_s^2-3)(c_s^2-1)-27T_{\Delta x}^{c_2}(c_s^2-3)(c_s^2-1)+T_{\Delta x}^{c_3}(c_s^2-9)(3c_s^2-1)+T_{\Delta x}^{c_4}(c_s^2-9)(3c_s^2-1))}{144} \\ \frac{c_s^2s_0(c_s^2-3)(3c_s^2-1)(3T_{\Delta x}^{c_1}-3T_{\Delta x}^{c_2}-T_{\Delta x}^{c_3}+T_{\Delta x}^{c_4})}{16} \\ \frac{c_s^2s_0(16T_{\Delta x}^{c_0}c_s^2(3c_s^4-10c_s^2+9)-9T_{\Delta x}^{c_1}(c_s^2-3)(3c_s^4-6c_s^2+1)-9T_{\Delta x}^{c_2}(c_s^2-3)(3c_s^4-6c_s^2+1)+T_{\Delta x}^{c_3}(3c_s^2-1)(c_s^4-18c_s^2+27)+T_{\Delta x}^{c_4}(3c_s^2-1)(c_s^4-18c_s^2+27))}{48} \end{bmatrix}$$

By the Eq. (??) can be shifted in time, we recursively over n timestep replace the shifted time function in it:

$$\underbrace{\mathbf{m}_{k,x}^{t+1} = \mathbf{P}\mathbf{m}_{k,x}^t + \mathbf{Q}\mathbf{m}_{k,x}^{eq,t}}_{\text{Time Shift}} \rightarrow \mathbf{m}_{k,x}^t = \mathbf{P}\mathbf{m}_{k,x}^{t-1} + \mathbf{Q}\mathbf{m}_{k,x}^{eq,t-1} \quad \text{Recursive Substitution } n \text{ times:} \quad \mathbf{m}_{k,x}^{t+1} = \mathbf{P}^n \mathbf{m}_{k,x}^{t-n+1} + \sum_{l=0}^{n-1} \mathbf{P}^l \mathbf{Q}\mathbf{m}_{k,x}^{eq,t-l} \quad (6.3)$$

The next steps follow in order to employ the Calay-Hamilton theorem [BGM22]. In this way, we need to find the characteristic polynomial $\mathcal{X}_P$ of the matrix $\mathbf{P}$ (the definition of characteristic polynomial are presented in toogle below), given by:

The characteristic polynomial of a square matrix ($P \in \mathbb{R}^{n \times n}$) (or $(\mathbb{C}^{n \times n})$) is defined as:

$$\boxed{\mathcal{X}_P(X) = \det(XI - P)} \quad \rightarrow \quad \mathcal{X}_P(X) = \sum_{i=0}^n X^i \gamma_i$$

where γ_i are the polynomial coefficients.

$$\mathcal{X}_P = \gamma_5 X^5 + \gamma_4 X^4 + \gamma_3 X^3 + \gamma_2 X^2 + \gamma_1 X^1 + \gamma_0,$$

where all the polynomial coefficients γ_i are shifted by $T_{\Delta x}^{c_0}$:

```
import pickle
from pathlib import Path
from sympy import symbols

X = symbols("X")
filename = Path("characteristic_polynomial.pkl")

if filename.exists():
    # Read the previously calculated result
    with filename.open("rb") as file:
        cp = pickle.load(file)

    print("Characteristic polynomial loaded from file.")
else:
    # Calculate only when the saved file does not exist
    cp = P.charpoly(X)

    with filename.open("wb") as file:
        pickle.dump(cp, file)

    print("Characteristic polynomial calculated and saved.")
```

Characteristic polynomial loaded from file.